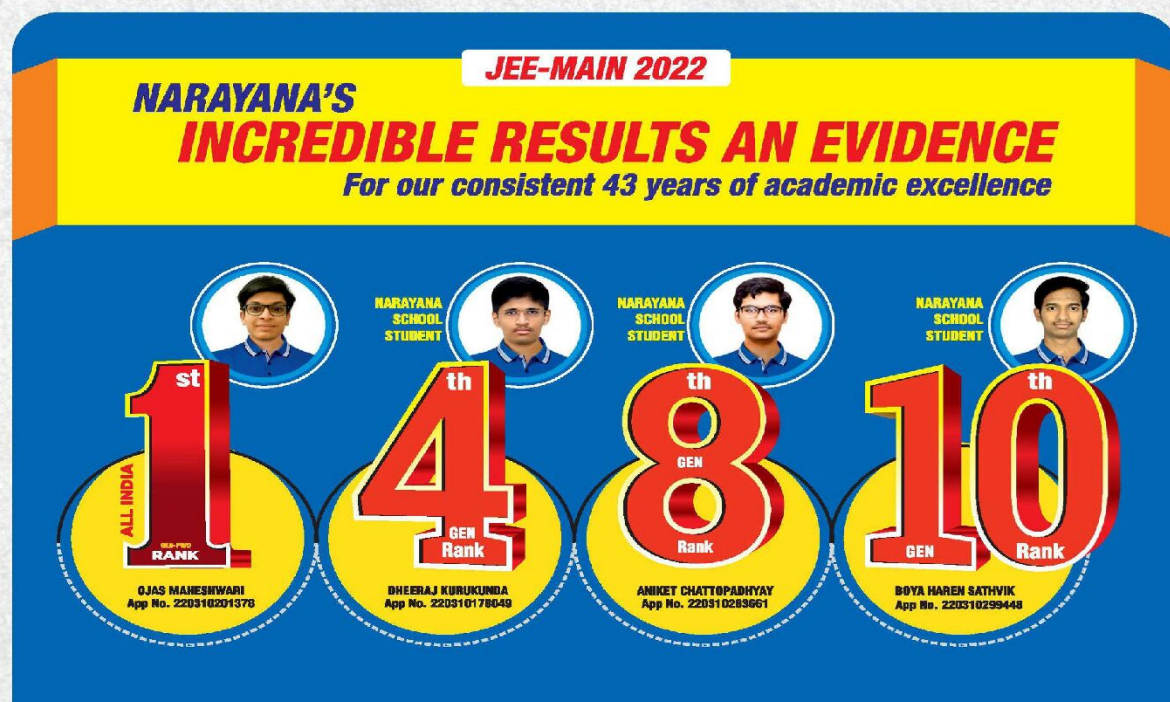




NARAYANA GRABS THE LION'S SHARE IN JEE-ADV.2022
EXCELLENCE IS OUR TRADITION 3402 RANKS OVERALL!



CHEMISTRY

ORGANIC CHEMISTRY PART-II(SR)

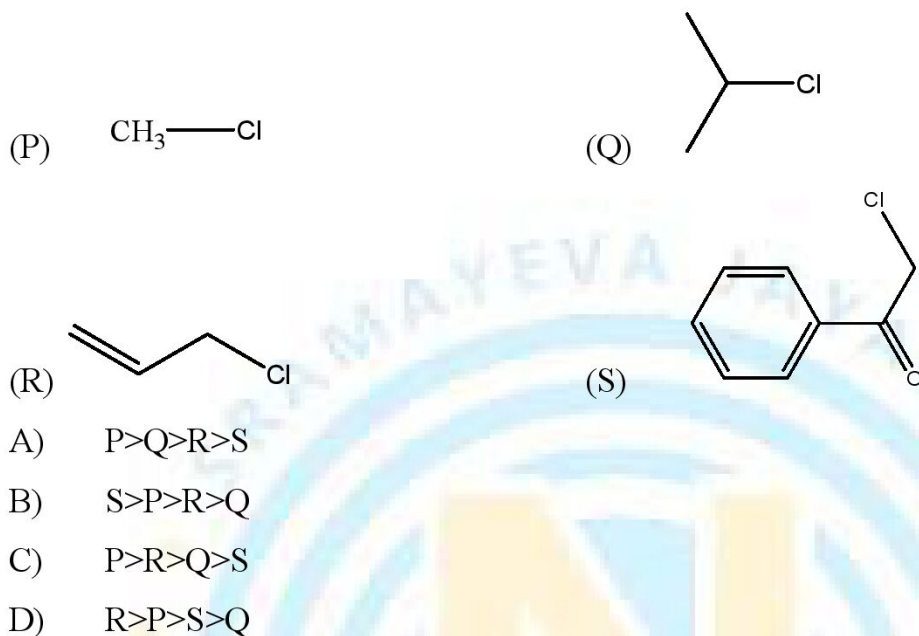
JEE MAIN IMPORTANT QUESTIONS



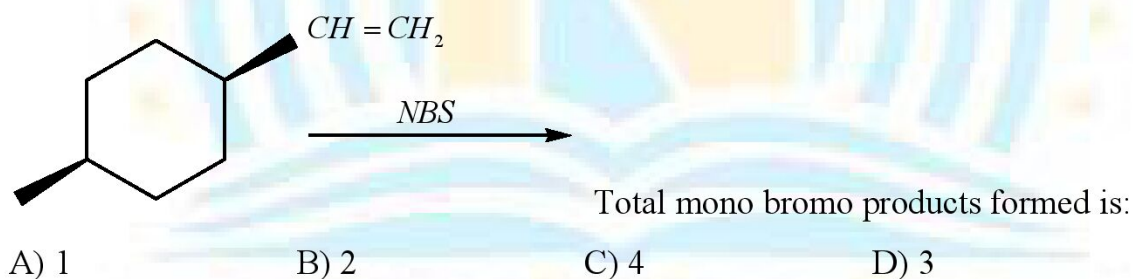
@bohring_bot

ORGANIC CHEMISTRY PART-II(SR)

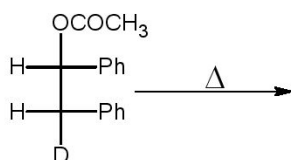
1. KI in acetone undergoes S_N2 reaction with each P,Q,R and S. The rates of the reaction vary as,



2.

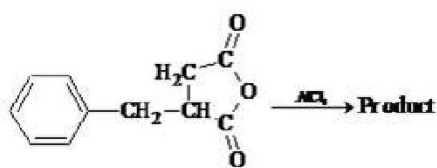


3. Identify the correct statement about the reaction;



- A) It is a syn elimination reaction and gives cis alkene
 B) It is an anti elimination reaction and gives trans alkene
 C) It is a syn elimination reaction and gives trans alkene
 D) The product does not contain deuterium

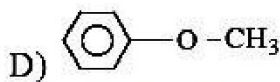
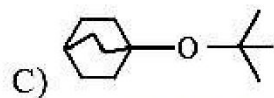
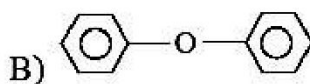
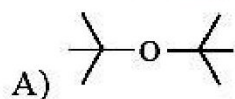
4. In the Friedel Craft's reaction given below



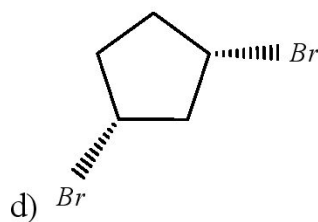
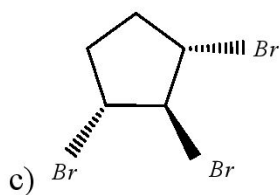
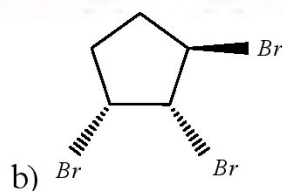
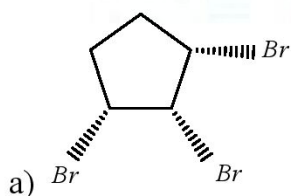
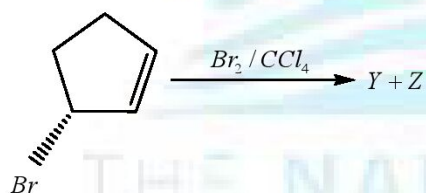
The major product is

- A)
- B)
- C)
- D)

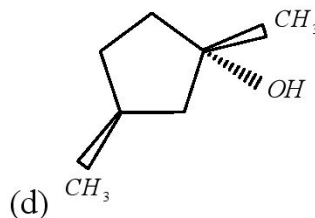
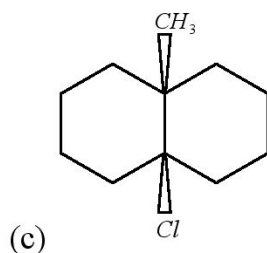
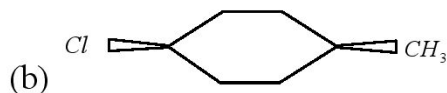
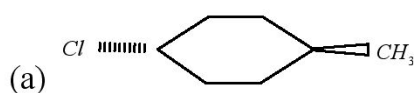
5. Which of the following ethers can be prepared by Williamson's Ether synthesis.



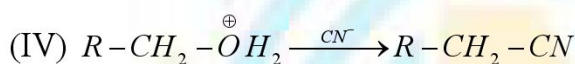
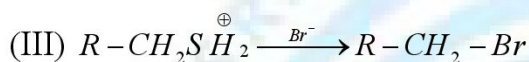
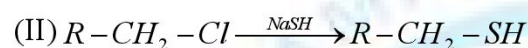
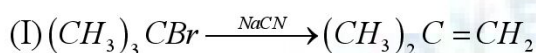
6. (R)-3 bromocyclopentene (shown below) reacts with Br_2 / CCl_4 to form two products, Y and Z, Y is not optically active. What is the structure of Y?



7. Which of the following on dehydro halogenation (using Alcoholic KOH)



8. Which of the following reaction feasible



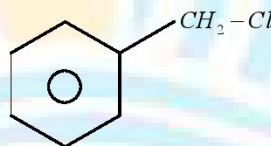
A) I, II, III & IV

B) II & III Only

C) I, II, III Only

D) II, III & IV Only

9. The IUPAC name of the following compound is



A) Chloromethyl benzene

B) ChloromethylPhene

C) Chlorophenyl methane

D) Benzyl chloride

10. Which of the following orders given below is True?

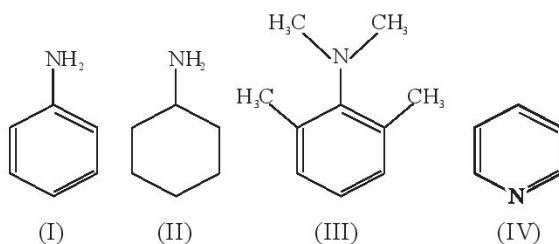
A) Bondlength (C-X) Order : $CH_3 > CH_3-Cl > CH_3Br > CH_3I$

B) Dipolemoment Order : $CH_3 > CH_3Cl > CH_3Br > CH_3I$

C) Bond enthalpy (C-X) order : $CH_3 > CH_3Cl > CH_3Br > CH_3I$

D) Dipolemoment Order: $CH_3I > CH_3Br > CH_3F > CH_3Cl$

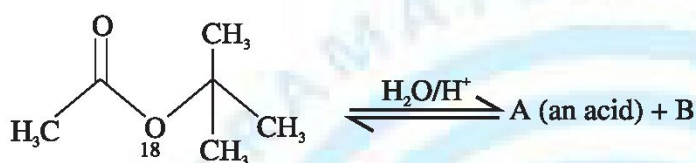
11. The C – N bond length in the molecules



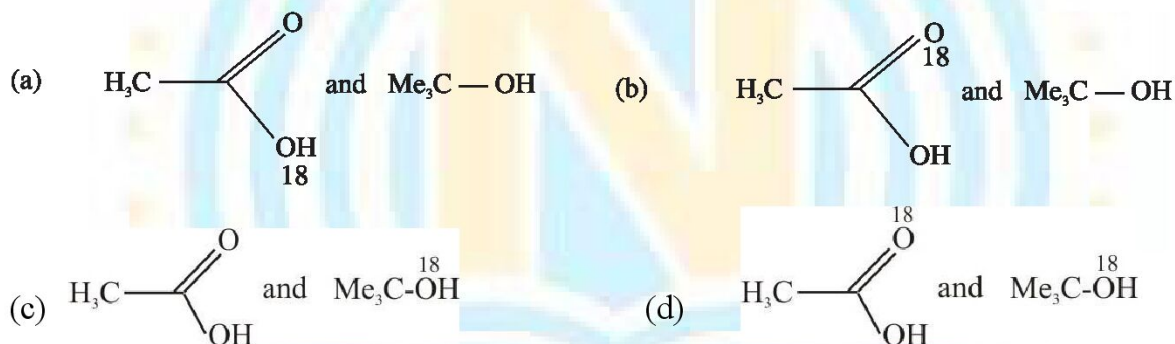
is most correctly in the order

A) II > III > I > IV B) II > III = I > IV C) III > IV > I > II D) IV > III > IV > I

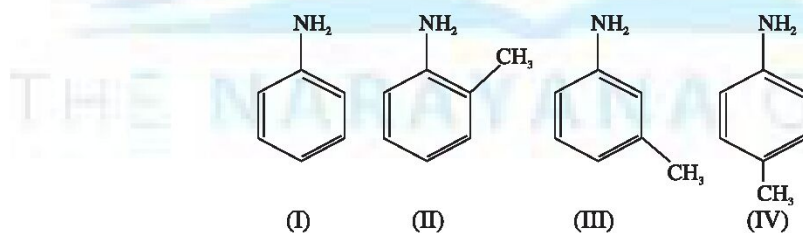
12.



Where A and B are, respectively,



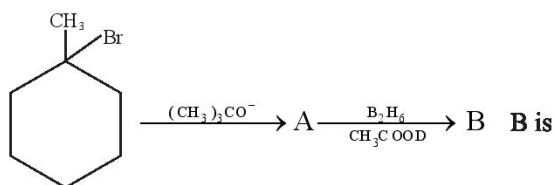
13. The pKa values of the conjugate acids of the amines

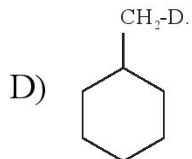
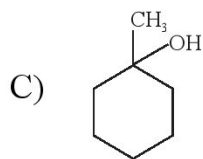
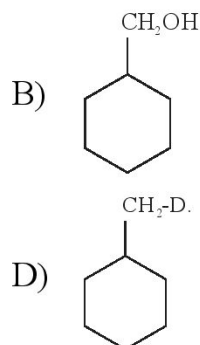
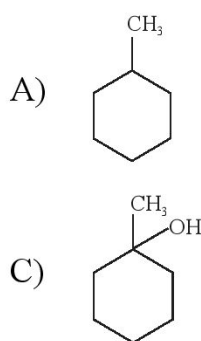


increases in the order:

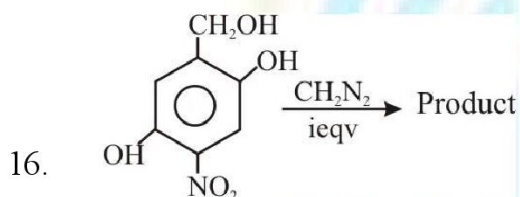
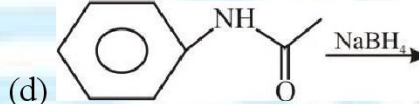
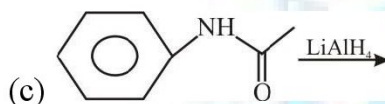
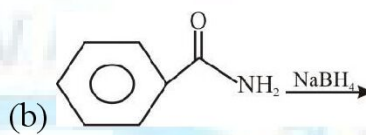
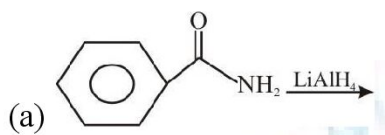
A) II, I, III, IV B) I, III, IV, II C) I, IV, III, II D) III, II, I, IV

14.

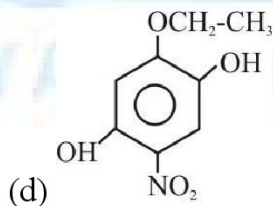
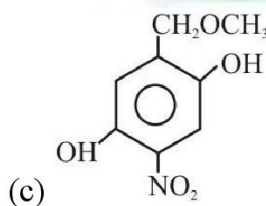
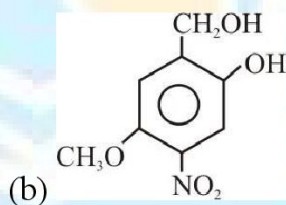
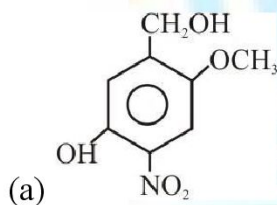




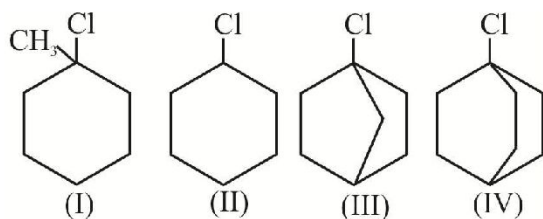
15. Find out the reaction in which obtained product given positive isocyanide test:



Product is



17. Arrange the following in decreasing order of hydrolysis



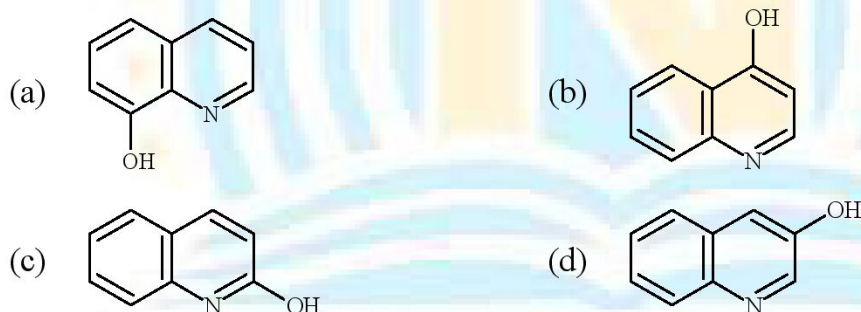
- (a) II > III > IV > I (b) I > II > IV > III
(c) II > I > IV > III (d) I > II > III > IV

18. Find the correct order of the solubility of the following alcohols in water:

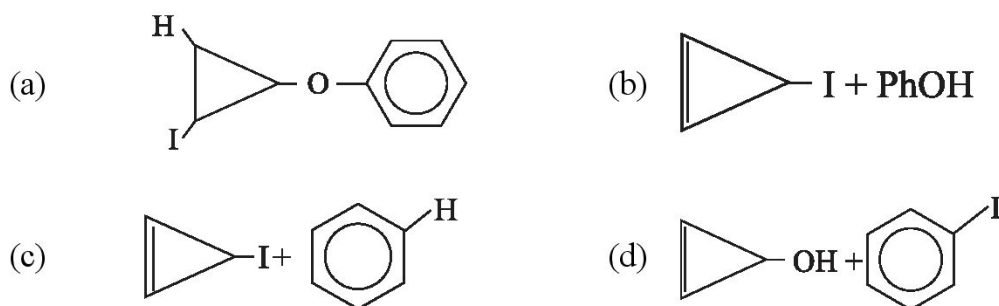


- (a) I > II > III > IV (b) IV > II > I > III
(c) I > III > II > IV (d) IV > III > II > I

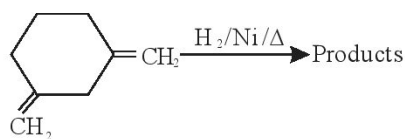
19. The compound having lowest boiling point is



20. Product. The product is



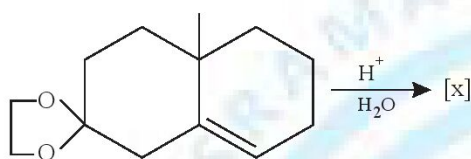
21.



The number of isomers obtained and number of fractions obtained upon fractional distillation of product mixture respectively are

- (a) 4 and 4 (b) 4 and 2 (c) 2 and 4 (d) 3 and 2

22.



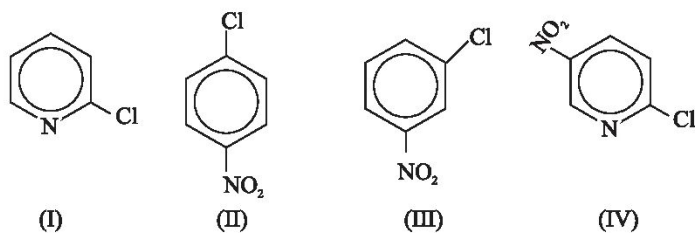
The end product [x] is

- (a) (b) (c) (d)

23. A polyhydric alcohol is acetylated by using ethanoyl chloride in presence of alkali and it was observed that molecular mass of alcohol is increased to 237% on acylation. If the molecular mass of polyhydric alcohol is 292, calculate the number of OH group present in the alcohol.

- (a) 2 (b) 3 (c) 4 (d) 5.

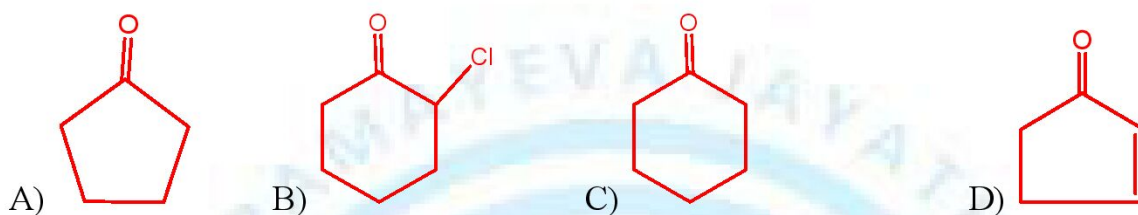
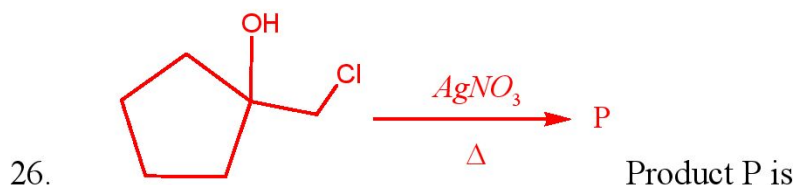
24. Arrange the following in decreasing order of nucleophilic substitution.



A) II > IV > I > III B) I > IV > II > III C) IV > I > II > III D) IV > II > I > III

25. Correct order of boiling point is

A) $EtCl > EtOH$ B) $EtOH > AcOH$ C) $EtOH > Me_2O$ D) $EtNH_2 > EtOH$



27. The ether in which all atoms exists in one plane

- A) Dimethyl ether
B) Methyl vinyl ether
C) Divinyl ether
D) Allyl ethyl ether

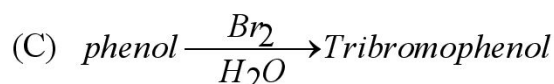


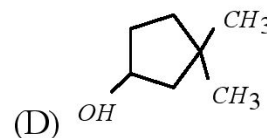
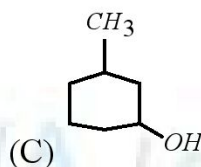
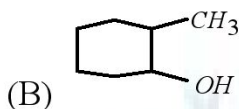
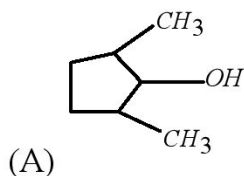
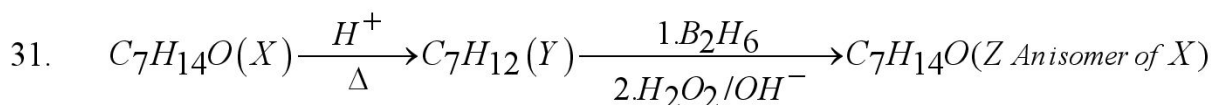
- A) HCHO, HCOOH B) HCHO, 2HCOOH
C) CO₂, 2HCOOH D) CO₂, HCHO, HCOOH

29. Which of the following compounds is **unlikely to react** with sodium metal?

- A) $C_2H_5OC_2H_5$ B) C_2H_5OH C) C_2H_5Br D) $C_2H_5NH_2$

30. Which of the following is not correctly matched



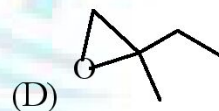
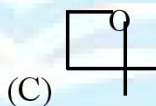
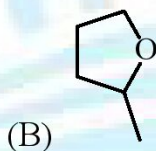
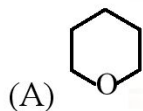


32. A chiral $\text{C}_5\text{H}_{10}\text{O}$ alcohol is reduced by catalytic hydrogenation to an achiral $\text{C}_5\text{H}_{12}\text{O}$ alcohol. The original alcohol is oxidized by activated MnO_2 to an achiral carbonyl compound ($\text{C}_5\text{H}_8\text{O}$)

Which of the following might be the chiral alcohol?

- (A) 1-penten-3-ol (B) 4-penten-2-ol (C) 3-methyl-2-buten-1-ol (D) 2-methyl-2-buten-1-ol

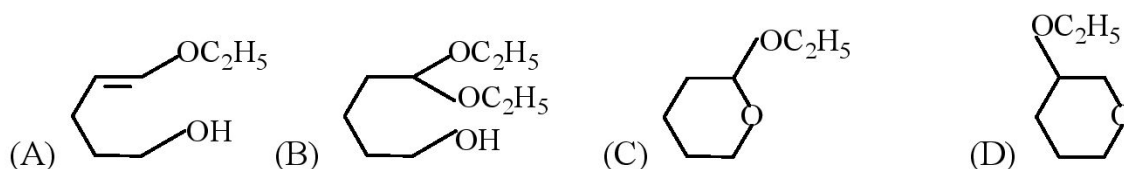
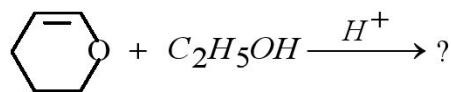
33. A chiral $\text{C}_5\text{H}_{10}\text{O}$ ether reacts with hot HI to give a $\text{C}_5\text{H}_{10}\text{I}_2$ product which on treatment with hot KOH in ethanol produces 1,3-pentadiene. What is the structure of the original ether?



34. A synthesis of 2,5-dimethyl-2-hexanol from 2-methylpropene requires the formation of two four-carbon intermediates, **X** and **Y**. These intermediates combine to give the desired product after the usual hydrolysis SSwork-up. Select appropriate methods of preparing **X** and **Y** from 2-methylpropene

- A) To get **X**, add HBr , then react with Mg in ether. To get **Y**, add water, acid-catalysis
 B) To get **X**, add HBr (peroxides), then react with Mg in ether. To get **Y**, react with $\text{C}_6\text{H}_5\text{CO}_3\text{H}$ in CH_2Cl_2
 C) To get **X**, add HOBr . To get **Y**, add B_2H_6 in ether, then NaOH
 D) To get **X**, add HOBr . To get **Y**, add HBr (peroxides), then react with Mg in ether

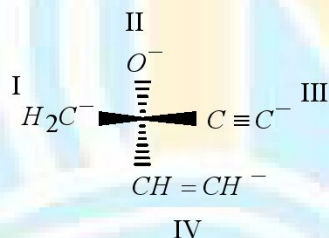
35. Which of the following is the product from ethanol addition to dihydropyran ?



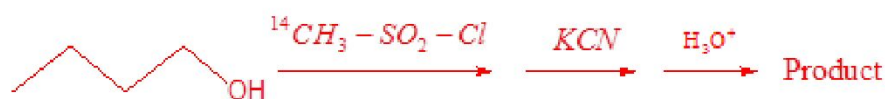
36. A $\text{C}_6\text{H}_{12}\text{O}$ compound does not react with Br_2 in CCl_4 , produces a flammable gas on treatment with LiAlH_4 , and reacts with H_2CrO_4 changing the color from orange to green.

Which of the following compounds best agrees with these facts?

- A) 1-methylcyclopentanol B) methoxycyclopentane
C) 2-cyclopropyl-2-propanol D) 2-cyclobutylethanol
37. Dehydrohalogenation is effected with strong bases. Alkoxides is one of the choice. The following species contain four basic sites. Which of the following is least preferred for protonation



- A) I B) II C) III D) IV

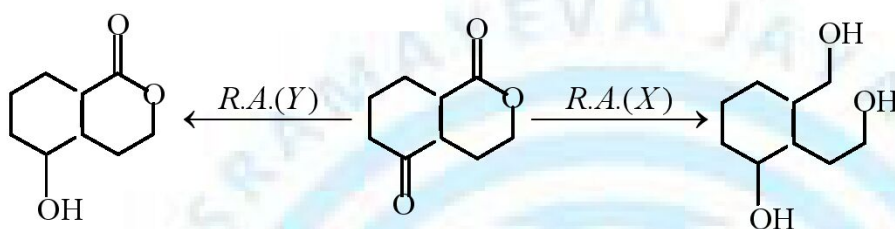


38. The product is



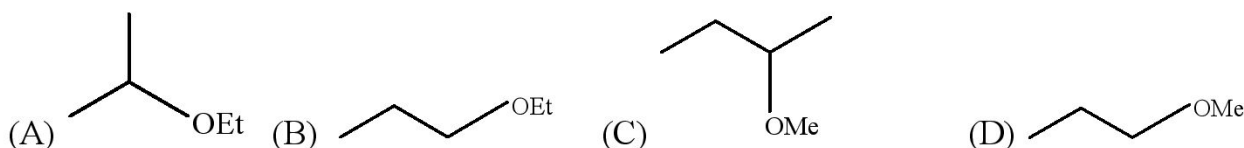
39. Generally primary alcohols are oxidised to carboxylic acids by acidified potassium dichromate, but with $\text{PCC}/\text{CH}_2\text{Cl}_2$ they are oxidised to aldehydes only. Because
- (A) $\text{PCC}/\text{CH}_2\text{Cl}_2$ is weak oxidising agent
 (B) Non-formation of gem diol in reaction mixture
 (C) With $\text{PCC}/\text{CH}_2\text{Cl}_2$, aldehyde formed is converted into non-oxidisable form
 (D) Aldehyde forms ester with oxidizing agent

40. The reducing agent (X and Y) in the following transformation are



- (A) X is NaBH_4 and Y is LiAlH_4 (B) X is H_2/Ni and Y is $\text{Na}-\text{EtOH}$
 (C) X is $\text{Zn(Hg)}/\text{Conc.HCl}$ and Y is H_2/Ni (D) X is LiAlH_4 and Y is NaBH_4
41. A water soluble $\text{C}_6\text{H}_{14}\text{O}_2$ compound is oxidized by lead tetraacetate (or periodic acid) to a single $\text{C}_3\text{H}_6\text{O}$ carbonyl compound. Which of the following would satisfy this fact
- (A) meso-2,3-dimethoxybutane (B) 1,2-diethoxyethane
 (C) meso-2,5-hexanediol (D) meso-3,4-hexanediol
42. 1-Propanol and 2-propanol can be best distinguished by
- A) Oxidation with KMnO_4 followed by reaction with Fehling solution
 B) Oxidation with acidic dichromate followed reaction with Fehling solution.
 C) Oxidation by heating with copper followed by reaction with Fehling solution
 D) Oxidation with concentrated H_2SO_4 followed by reaction with Fehling solution

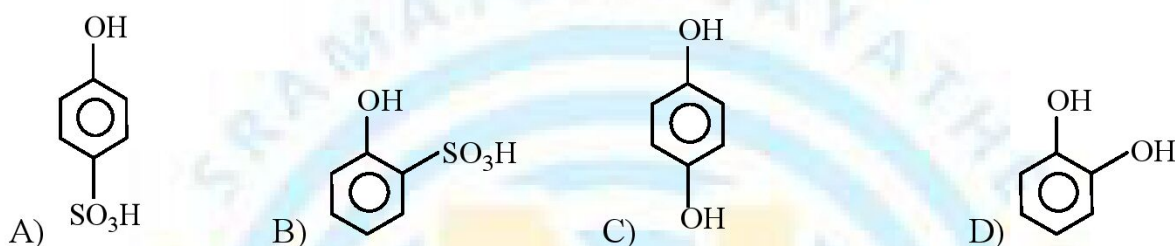
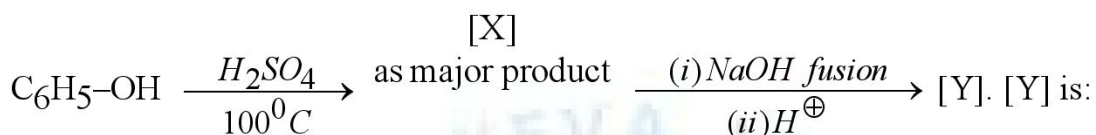
43. $\text{X}(\text{C}_5\text{H}_{12}\text{O}) \xrightarrow[\text{(ii) KOH}]{\text{(i) PCl}_5} \text{Y} + \text{Z}$, both Y and Z are alcohols and both can give positive haloform test. The compound X is



44. An organic compound with molecular formula C_7H_8O dissolves in $NaOH$ and gives characteristic colour with $FeCl_3$. On treatment with Br_2 it gives tribromo product $C_7H_5OBr_3$. The compound is

A) O-cresol
B) Anisol
C) m-cresol
D) p-cresol

45. In the given reaction



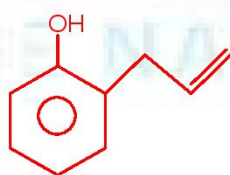
46. $Ph-SO_3H \xrightarrow{NaOH} A \xrightarrow[Fuse]{NaOH} B \xrightarrow{H_3O^\oplus} C$ Pdt C is

A) $Ph-OH$ B) $Ph-H$ C) $Ph-O^-Na^+$ D) $Ph-SO_3^-Na^+$

47. $PhOH \xrightarrow{NaOH} A \xrightarrow{\text{allyl bromide}} B \xrightarrow{\Delta} C + D$ Identify the wrong match

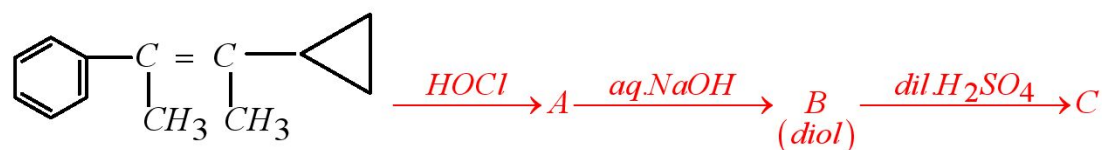
A) A is $Ph-O^-Na^+$

B) B is 

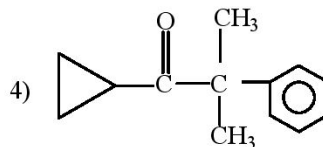
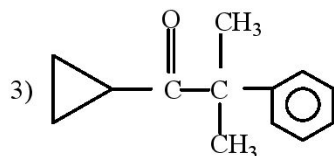
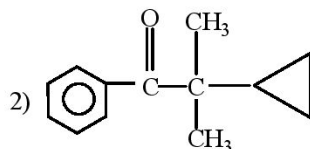
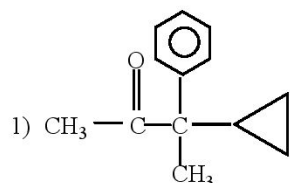
C) C Can be 

D) Conversion B to (C+D) Involve Carbocation intermediate

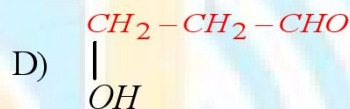
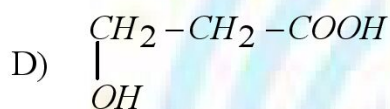
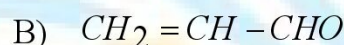
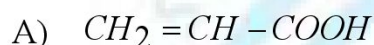
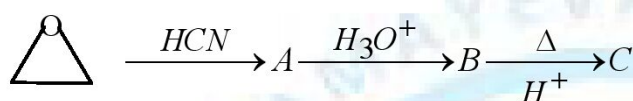
- 48.



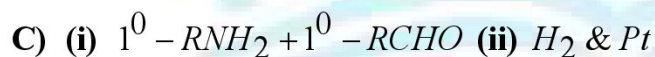
The product C is:



49. The structures of C



50. What set of conditions would be useful for preparing a 2°-amine



51. When Compound $\text{BrCH}_2\text{CH}_2\overset{\text{O}}{\parallel}\text{COC}(\text{CH}_3)_3$ is subjected to acid hydrolysis, Compound

$\text{BrCH}_2\text{CH}_2\overset{\text{O}}{\parallel}\text{COH}$ is regenerated and 2-methylpropene is formed. Which of the following transformations accounts for the formation of 2-methylpropene?

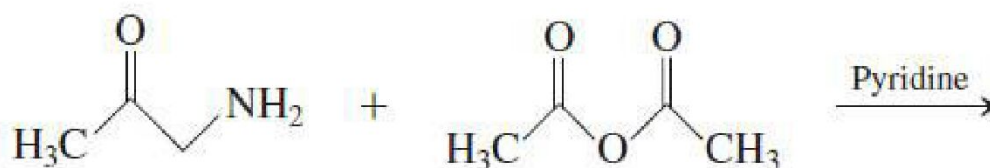
A) Dehydration of an alcohol

B) Nucleophilic displacement of a halide

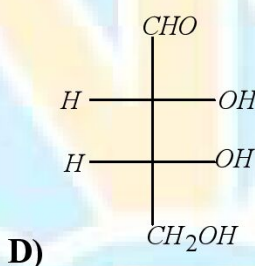
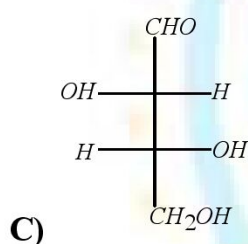
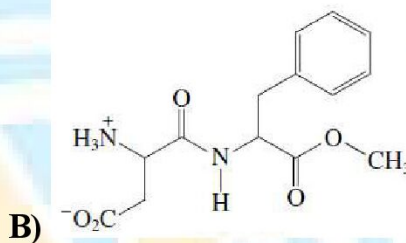
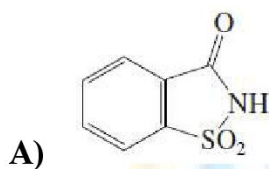
C) Cleavage of an ether

D) Oxidation of an acid

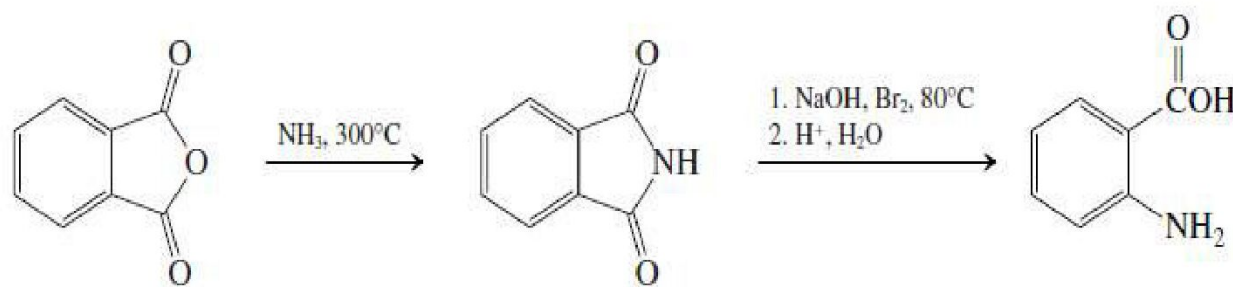
52. Which of the following functional groups are present in the product



- A) Two ketones and an amide B) A ketone and an amide
C) A ketone and an ester D) An ester and an amide
53. HNO_3 oxidizes primary alcohols and aldehydes to the corresponding carboxylic acid. One of the sweeteners was found to be optically active; however, its optical activity vanished upon treatment with HNO_3 . Which of the following compounds could it be?



54. Which among following possess relatively low boiling point
- A) $CH_3CH_2CH_2CH_2COOH$ B) $(CH_3CH_2CH_2)_2O$
C) $CH_3(CH_2)_5NH_2$ D) $CH_3(CH_2)_5OH$
55. 2-Aminobenzoic (anthranilic) acid is prepared from 1,2-benzenedicarboxylic anhydride (phthalic anhydride) by using the two reactions shown here. Which one of the following reactions is associated with this transformation.

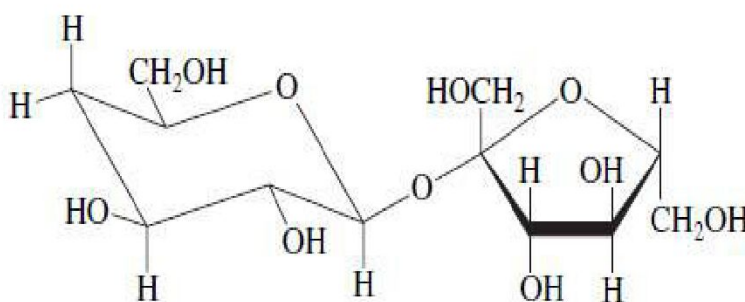


- A) Gabriel phthalimide synthesis B) Perkin reaction
 C) Hoffman bromamide degradation D) Claisen reaction

56. A positive carbylamine test is given by:

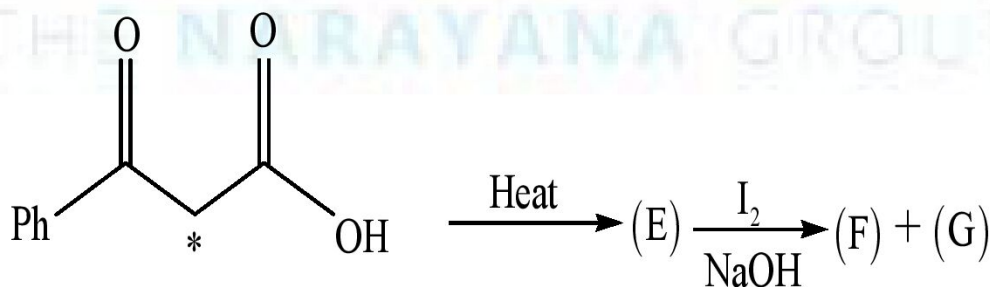
- A) N, N – Dimethyl aniline B) N – Methyl – o – Methyl aniline
 C) p – Methyl benzylamine D) All of these

57. One of the statements below is correct about the sugar shown. Which one?

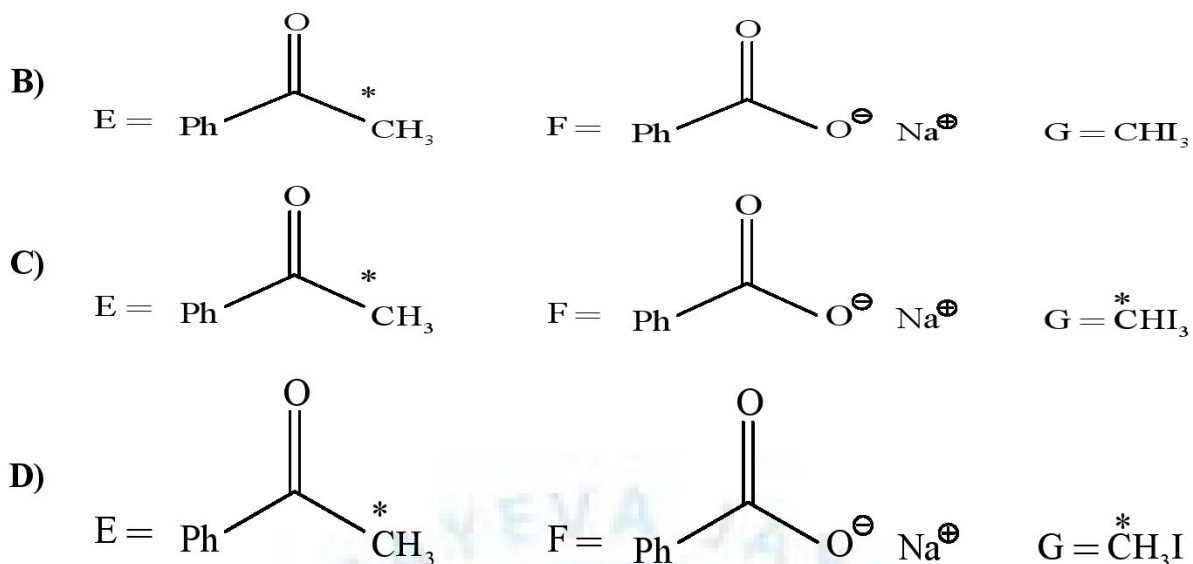


- A) It is a nonreducing sugar B) It forms an osazone
 C) It exists in two anomeric forms D) It undergoes mutarotation

58. In the following reaction sequence, the correct structure of E, F and G are (* implies C^{13} labelled carbon)



- A) $\text{E} = \text{Ph}-\text{C}(=\text{O})-\text{C}^*(\text{H})-\text{CH}_3$ $\text{F} = \text{Ph}-\text{C}(=\text{O})-\text{O}^\ominus \text{Na}^\oplus$ $\text{G} = \text{CHI}_3$



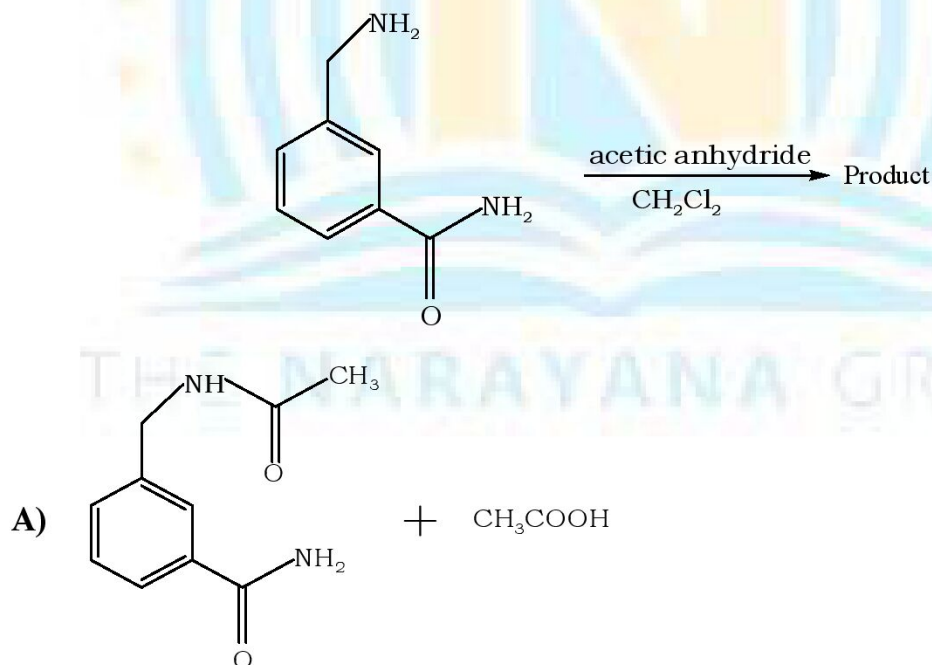
59. Which reagent could be used to separate a mixture of aniline and toluene?

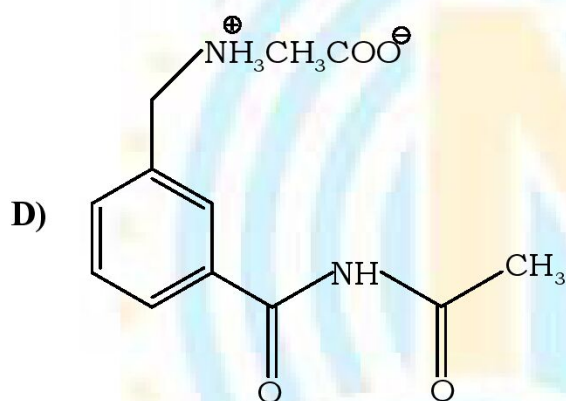
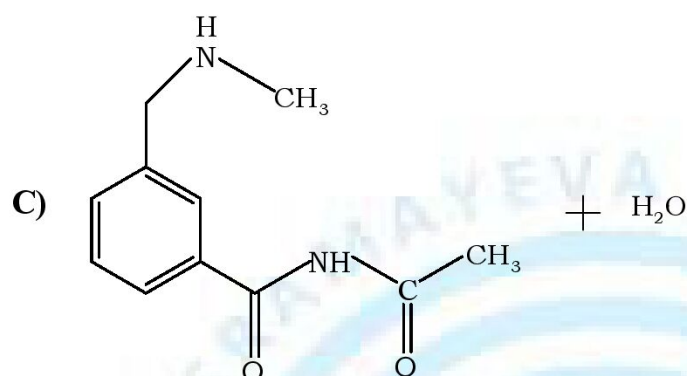
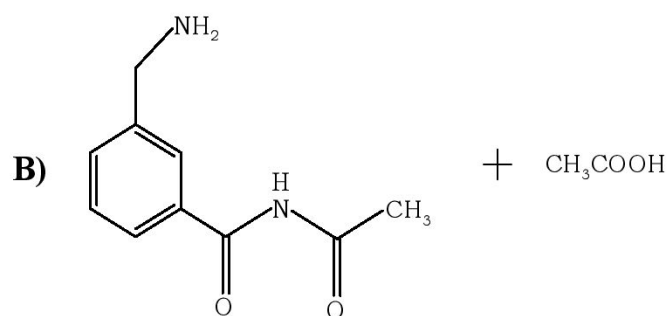
- A) KMnO_4 in H_2O B) Dilute NaOH C) Dilute NaHCO_3 D) Dilute HCl

60. Vitamin B_1 is

- A) Riboflavin B) Cobalamine C) Thiamine D) Pyridoxine

61. In the reaction shown below, the major product formed is

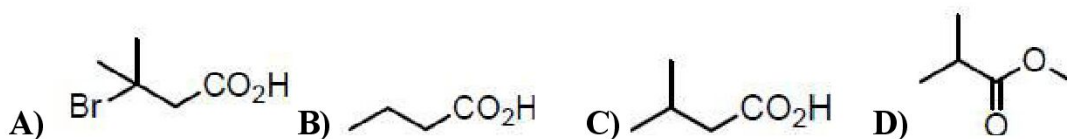
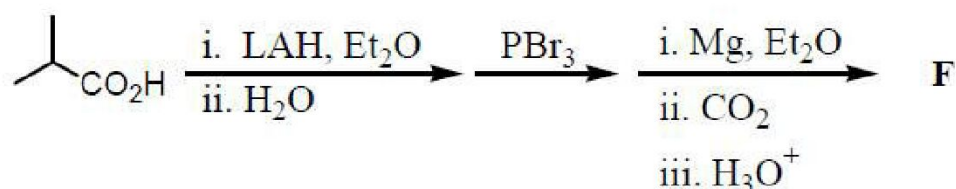




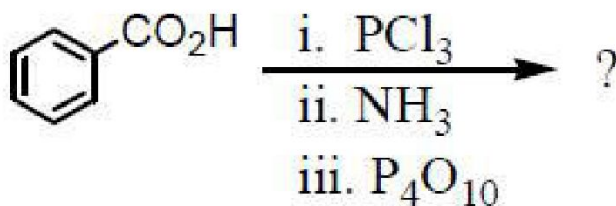
62. Which reagent would best serve as the basis for a simple chemical test to distinguish between $\text{C}_6\text{H}_5\text{CH}=\text{CHCOOH}$ and $\text{C}_6\text{H}_5\text{CH}=\text{CHCH}_3$?

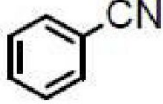
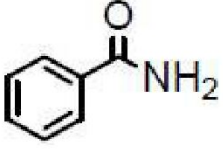
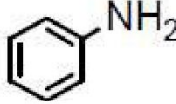
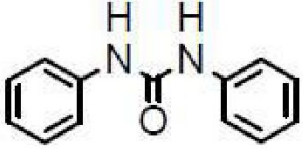
A) $\text{KMnO}_4 / \text{H}_2\text{O}$ B) $\text{Br}_2 / \text{CCl}_4$ C) $\text{CrO}_3 / \text{H}_2\text{SO}_4$ D) $\text{NaHCO}_3 / \text{H}_2\text{O}$

63. What would be the final product, **F**, of the following sequence of reactions?

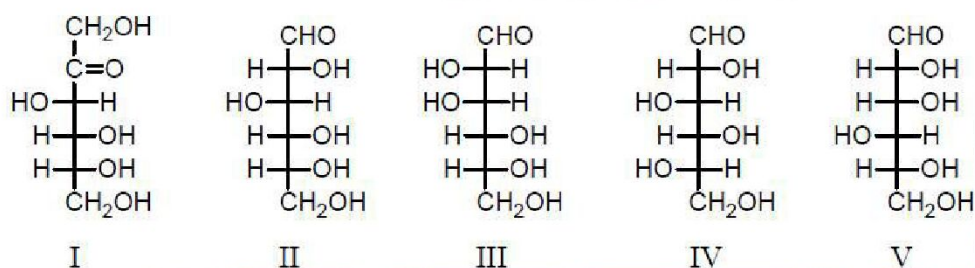


64. What would be the final product of this reaction sequence?

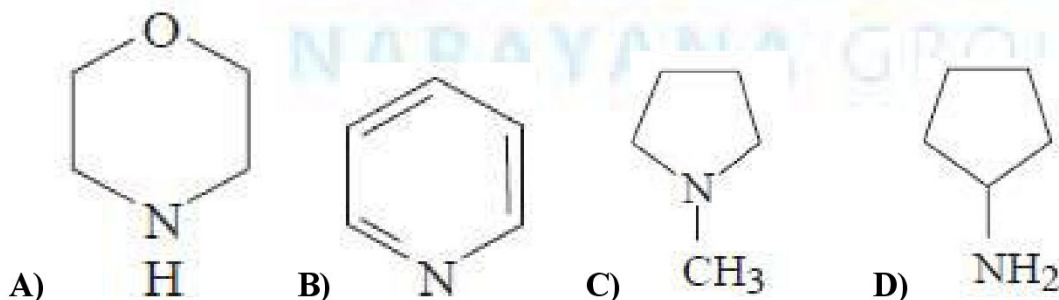


- A)  B)  C)  D) 

65. Sugars that would yield the same phenylosazone are:



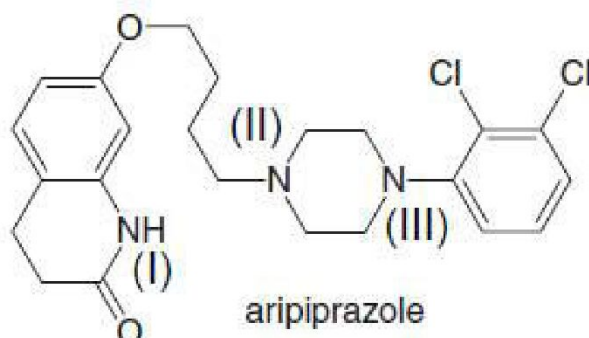
- A) I, II and V B) I, II and III C) II and III D) III and IV
66. Which is incorrect statement?
- A) Insulin is a protein which contains 51 amino acids
- B) Most naturally occurring amino acid have L – configuration
- C) Vitamins A, D, E and K cannot be stored in our body
- D) Vitamin B and C are water soluble
67. One of the following is often used to prepare enamines from aldehydes and ketones. The others do not yield enamines. Identify the enamine-forming compound



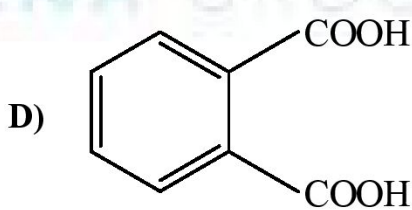
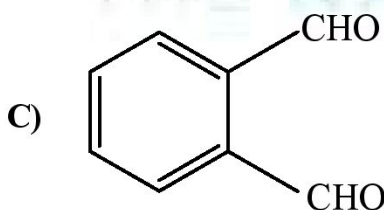
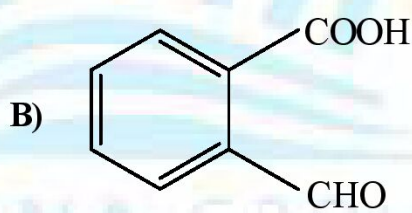
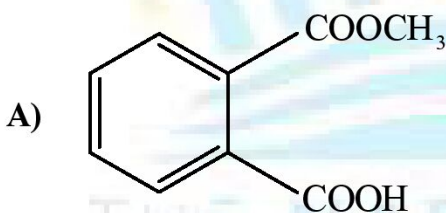
68. What reagent would be best for converting the amide of (R)-2-phenylpropanoic acid, $\text{C}_6\text{H}_5\text{CH}(\text{CH}_3)\text{CONH}_2$ into (R)-1-phenylethylamine?
- A) excess H_2 & Pt B) NaOBr in aqueous base

C) NaBH_4 in methanolD) LiAlH_4 in ether

69. Ability is the trade name for aripiprazole, a drug used to treat depression, schizophrenia, and bipolar disorders. Rank the N atoms in aripiprazole in order of increasing basicity

A) $\text{I} > \text{II} > \text{III}$ B) $\text{III} > \text{II} > \text{I}$ C) $\text{II} > \text{III} > \text{I}$ D) $\text{II} > \text{I} > \text{III}$

70. Which of the following reactants on reaction with conc NaOH followed by acidification gives the following lactone as the only product?



71. The carboxyl functional group ($-\text{COOH}$) is present in

A) Picric acid

B) Barbituric acid

C) Aspirin

D) Both (B) and (C)

72. The two form of D – gulcopyranose obtained from the solution of D – Glucose are called

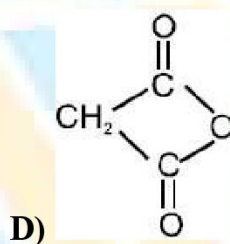
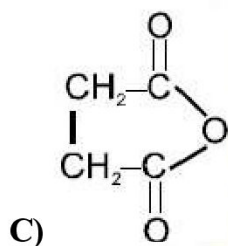
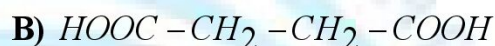
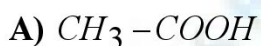
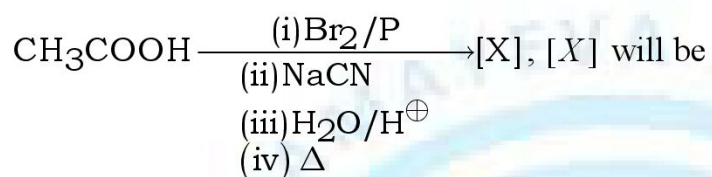
A) isomers

B) Anomers

C) Enantiomers

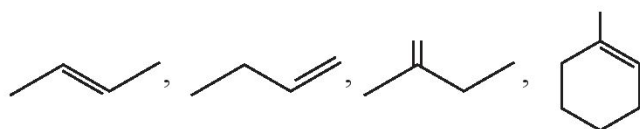
D) Conformers

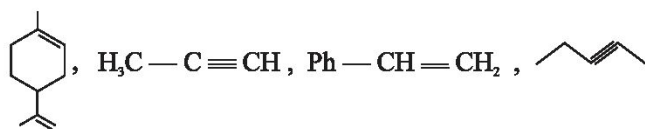
73. Which one of the following compounds does not give positive test for nitrogen in the Lassaigne's test?
 A) Urea B) Hydroxyl amine C) Glycine D) Diethylamine
74. The sequence in which amino acids are linked to one another in a protein molecule is called its:
 A) Primary structure B) Secondary structure
 C) Tertiary structure D) Quaternary structure
75. In the given reaction



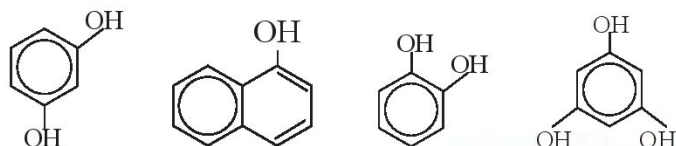
NUMERICAL

76. N-N bond order for azide ion in its most stable canonical structure is.
77. The number of the bromine atoms present in the product obtained from the reaction of phenol with bromine in the presence of CS_2 .
78. The number of carbon atoms present in the drug "paracetamol".
79. 10 mL of 1mM surfactant solution forms a monolayer covering 1.5 cm^2 on a polar substrate. If the polar head is approximately as cube, what is its edge length in pm?
80. A bottle of H_2O_2 is labelled as "10V" H_2O_2 . 336 mL of this solution of H_2O_2 is titrated against 0.04 M acidified solution of KMnO_4 . The volume of KMnO_4 in litre is
81. Of the following compounds given below find how many compounds will produce CO_2 on oxidative ozonolysis.



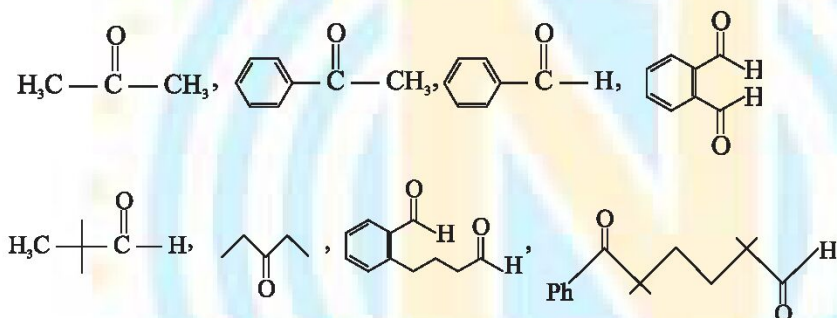


82. Among the following how many can reduce tollen's reagent
 HCOOH , CH_3COCH_3 , PhCOOH , CH_3CHO , PhCOCH_3 ,



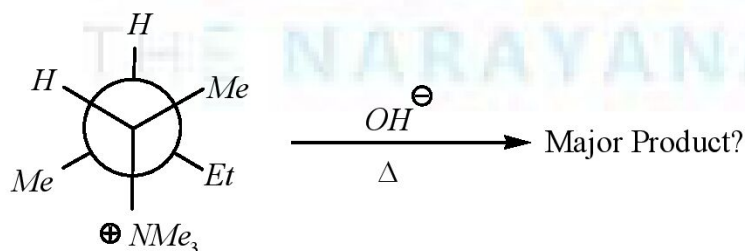
83. Calculate the number of lone pairs in guanine (present in DNA) which are involved in resonance.

84. Of the following carbonyl compounds, how many would give aldol condensation reaction?



85. Of the following compounds how many can liberate CO_2 from NaHCO_3 ?
 Picric acid, benzene sulfonic acid, formic acid, nitric acid, p-nitro phenol, glucose.

86.



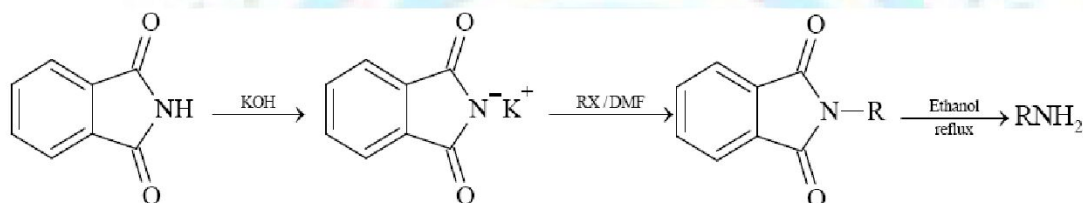
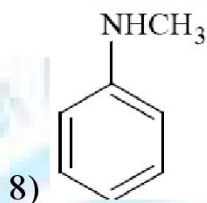
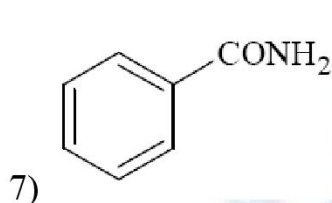
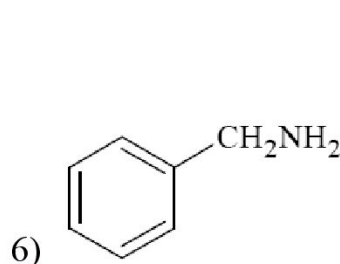
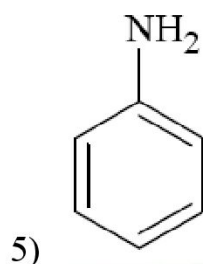
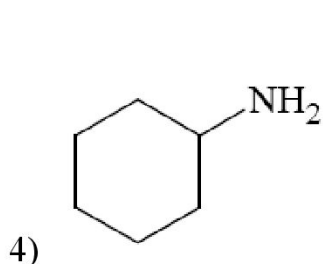
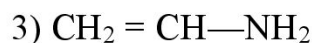
The number of hyper conjugative structures possible for the alkene(Major) obtained in the above reaction is

87. How many stereocenters are present in the small, naturally occurring protein
 glycylalanylalanine (Gly – Ala – Ala)

A) Zero B) one C) two D) three

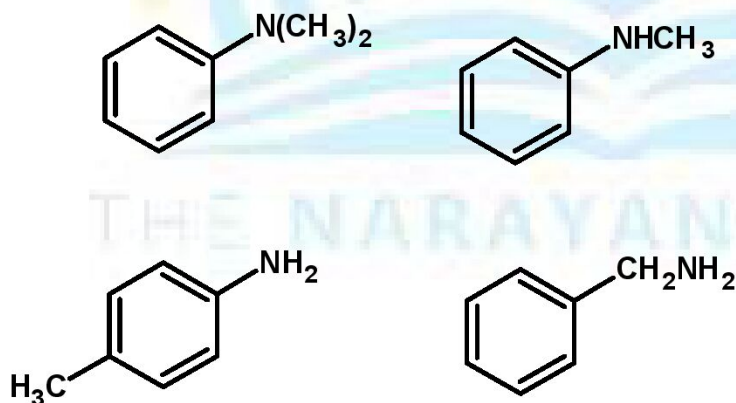
88.

Out of the given amines, how many cannot be prepared by this method

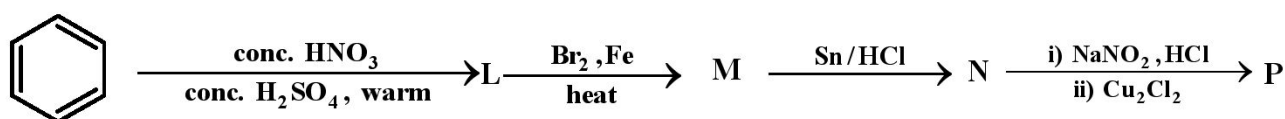


89.

Amongst the compounds given below, how many of them form a brilliant colored dye on treatment with NaNO_2 in dil. HCl , followed by addition to an alkaline solution of β -naphthol is/are ?



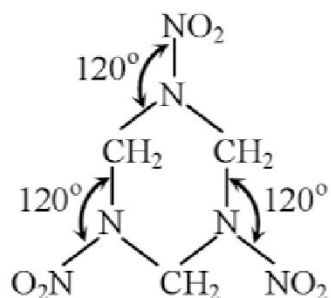
90. The product **P** obtained through the following sequence of reactions



According to **IUPAC** nomenclature, what position will be assign to **chlorine** atom from 1 to 6 in product **P**

91. The diagram shows the structure of **RDX**, one of the main component of *Semtex*(a widely used explosive).

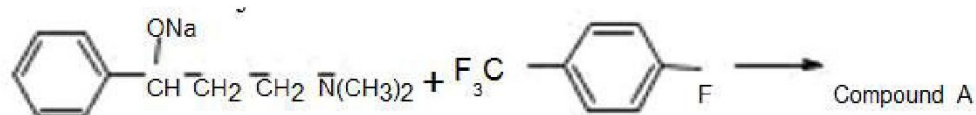
How many stereoisomers does **RDX** have ?



92. The spots of coloured compounds are visible on TLC plate due to their original colour. The spots of colourless compounds, which are invisible to the eye but fluoresce, can be detected by putting the plate under ultraviolet light. Sometimes an appropriate reagent may also be sprayed on the plate. For example, amino acids may be detected by spraying the plate with ninhydrin solution. No of sp^2 carbon atoms in ninhydrin is.....
93. It required 0.70 gm of hydrocarbon (A) to react completely with 2 gm Br_2 . On treatment of (A) with HBr, it yielded monobromoalkane (B). The same compound (B) was obtained when (A) was treated with HBr in the presence of peroxide. How many number of carbon atoms that are present in alkane(A)
94. In how many of the following pairs the former is more reactive than later with sodium methoxide in methanol at $50^\circ C$?

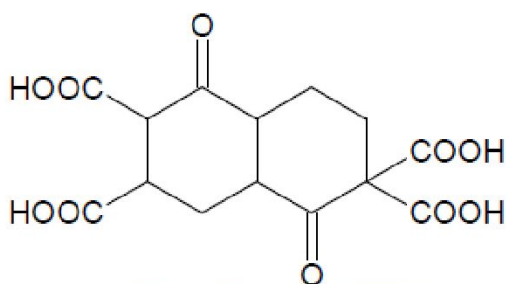
- (i) Chlorobenzene and o-chloronitrobenzene
- (ii) p-chloronitrobenzene and m-chloronitrobenzene
- (iii) 4-chloro-3-nitroacetophenone and 4-chloro-3-nitrotoluene
- (iv) 2-fluoro-1,3-dinitrobenzene and 2-chloro-1,3-dinitrobenzene
- (v) 4-chloronitrosobenzene and 4-chloroanisole
- (vi) 2-iodo-1-(trifluoromethyl)benzene and 2-fluoro-1-(trifluoromethyl)benzene

95. An article in the journal of chemical education described the following reaction

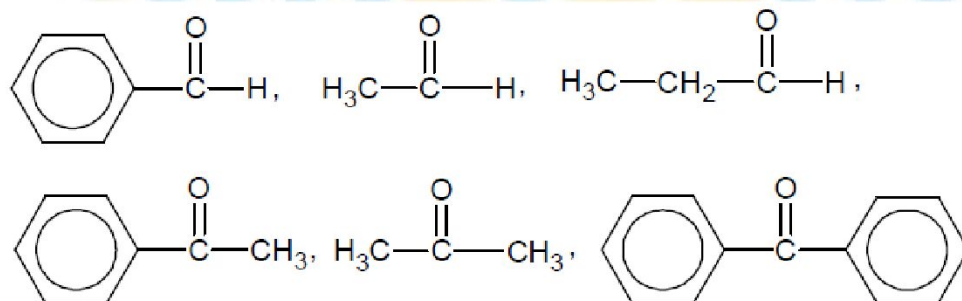


Fluoxetine hydrochloride (Prozac) is a widely prescribed antidepressant drug. It differs from compound 'A' in having an -NHCH_3 group in place of $\text{-N(CH}_3)_2$. If the number of hydrogens in Prozac is 'X', the value of X is ?

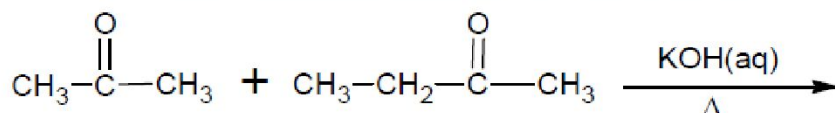
96. How many moles of CO_2 will be released when following compound is heated



97. Among the following the number of compounds which react with Fehling solution is

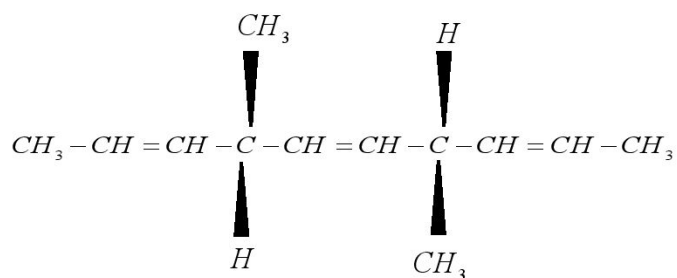


- 98.

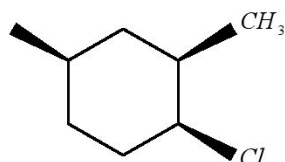


Calculate total number of aldol condensation product (s) including stereoisomers.

99. The number of optically active products obtained from the complete ozonolysis of the given compound is:



100.

Total number of $\text{S}_{\text{N}}1$ products of given compound are

KEY AND HINTS

SINGLE ANSWER QUESTIONS

1. Ans: B

Sol:

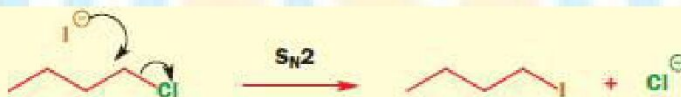
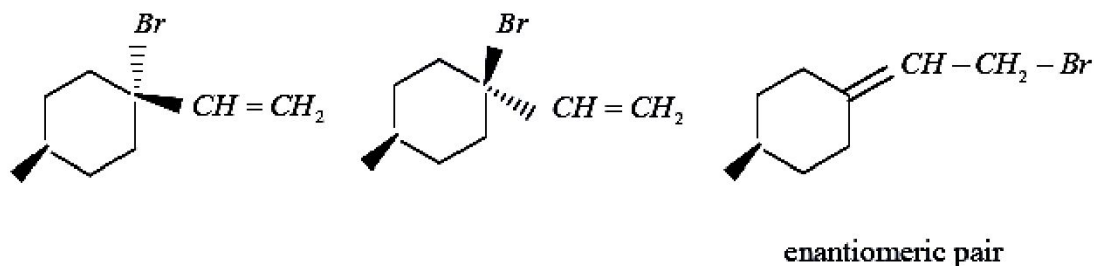


Table 17.10 Relative rates of $\text{S}_{\text{N}}2$ reactions of alkyl chlorides with the iodide ion

Alkyl chloride	Relative rate	Comments
$\text{Me}-\text{Cl}$	200	least hindered alkyl chloride
	0.02	secondary alkyl chloride; slow because of steric hindrance
	79	allyl chloride accelerated by π conjugation in transition state
	200	benzyl chloride slightly more reactive than allyl; benzene ring better at π conjugation than isolated double bond
	920	conjugation with oxygen lone pair accelerates reaction
	100 000	conjugation with carbonyl group much more effective than with simple alkene or benzene ring. These α -carbonyl halides are the most reactive of all

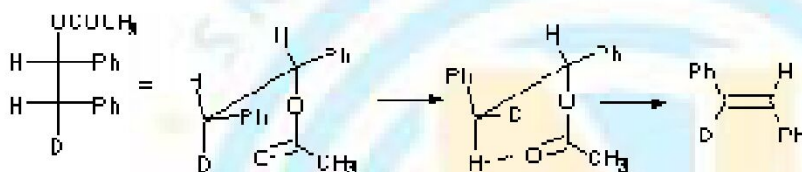
2. Ans: C

Sol:



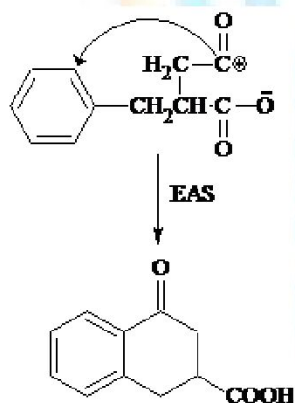
3. Ans: C

Sol:



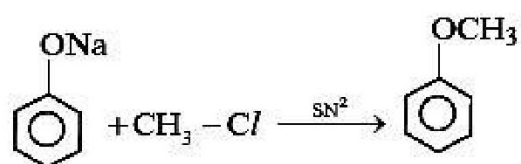
4. Ans: D

Sol:



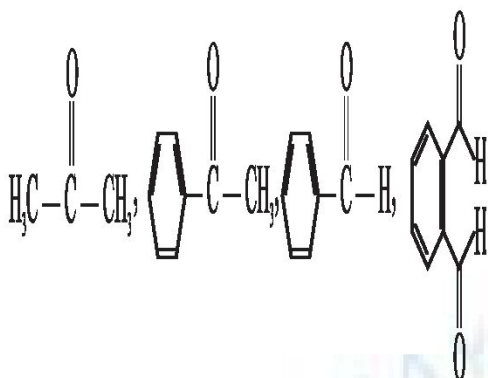
5. Ans: D

Sol:



6. Ans: C

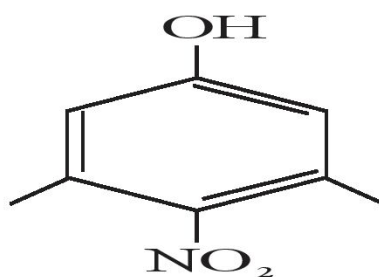
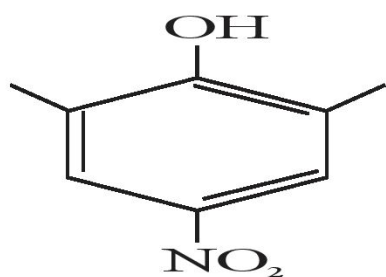
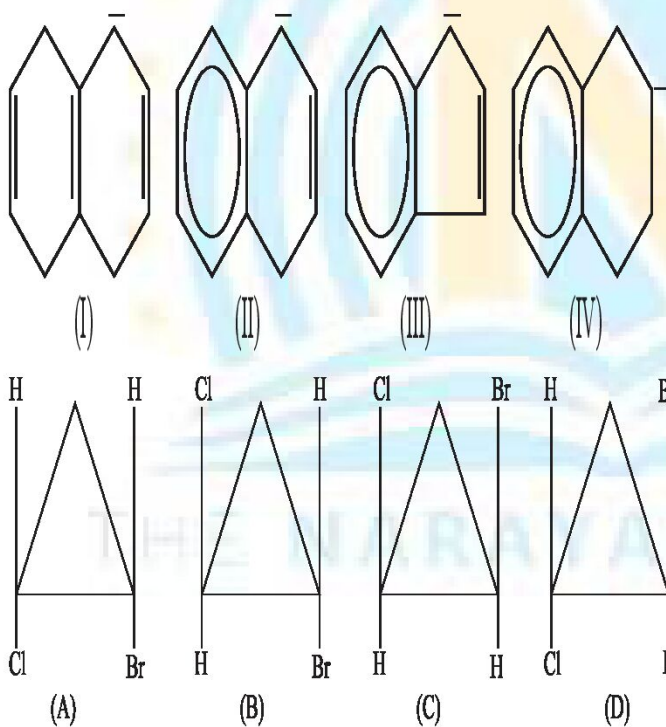
Sol:



Anti addition of halogen takes place on alkene.

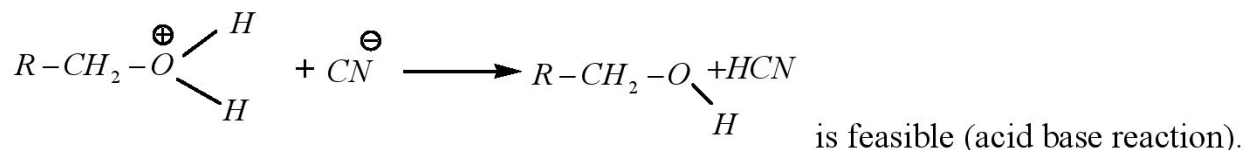
7. Ans: D

Sol:



8. Ans: C

Sol:



9. Ans: C

Sol: Refer NCERT Text book – class-XII (Page No-28D)

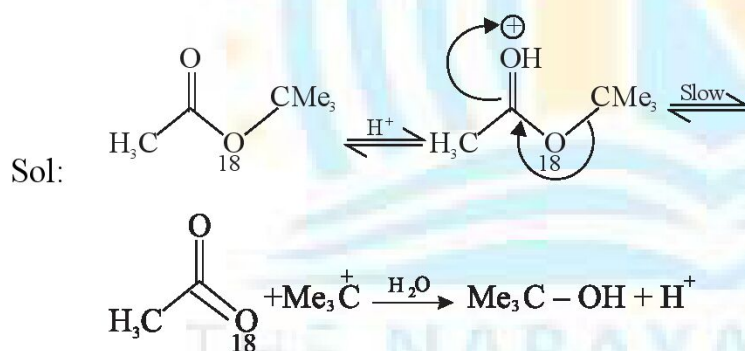
10. Ans: C

Sol: Refer NCERT- Text Book Class-XII(Part-II-Page No-286)

11. Ans: A

Sol: As the s-character of the hybridized orbital of C-atom increases, the bond length decreases. A double bond is shorter than a single bond. Participation of lone pair of N atom in resonance induces partial double-bond character between C and N.

12. Ans: B

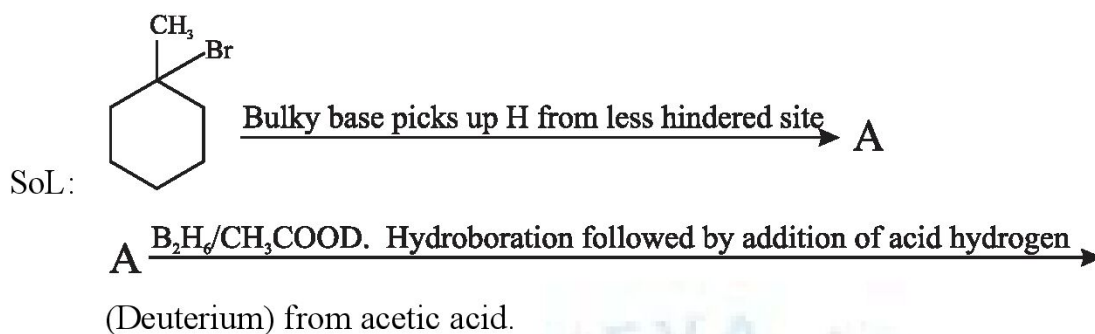


The above mechanism of ester hydrolysis is known as alkyl-oxygen cleavage unimolecular.

13. Ans: A

Sol: Due to ortho effect ortho isomer has the lowest basic strength and the highest acidic strength of its conjugate acid and hence lowest pKa. From m-position $-CH_3$ only exerts +I effect and from p-position it exerts +I as well as hyper conjugation effects.

14. Ans : (d)



15. Ans.(a)

SoL: Isocyanide test is given by 1° Amine.

16. Ans.(b)

Sol: Diazomethane will abstract the most acidic hydrogen

17. Ans.(b)

Sol: Hydrolysis involves $\text{S}_{\text{N}}1$ mechanisms, formation of carbocation is difficult at bridge head position.

18. Ans: (d)

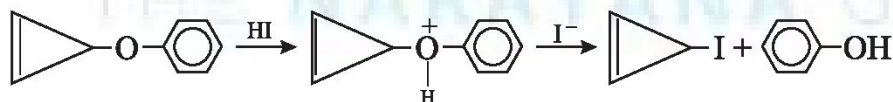
Sol: The lesser is the surface area of hydrophobic alkyl part higher is the solubility.

19. Ans:(a)

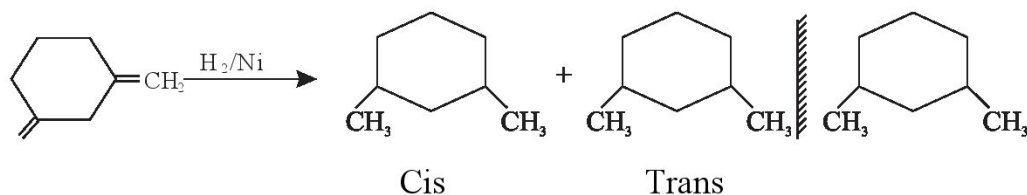
Sol: Develops five membered intramolecular H-bonding leading to lower bpt.

20. Ans :

Sol:



21. Ans: (d)

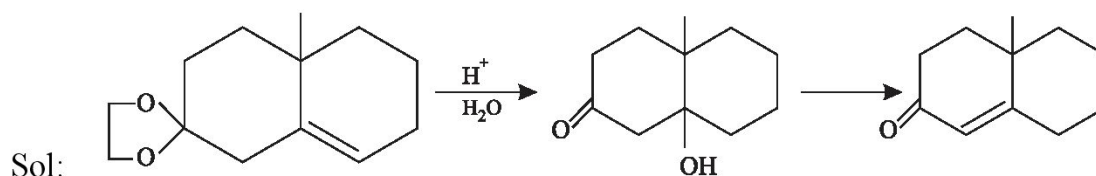


SoL:

Total isomer will be 3 and only 2 product will be obtained in fractional distillation d and l

separated together.

22. Ans: (d)



23. Ans: (b)

Sol: $\text{ROH} \rightarrow \text{ROCOCH}_3$

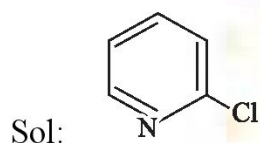
H replaced by COCH_3 i.e. mass increased by 42.

$$\text{Mass of acylated product} = \frac{92 \times 237}{100} = 218$$

$$\text{Increase in mass} = 218 - 92 = 126$$

$$\therefore \text{No. of OH group} = \frac{126}{42} = 3$$

24. Ans: C



is more reactive towards nucleophilic substitution than p-chloronitrobenzene. Presence of NO_2 group increases the reactivity towards nucleophilic substitution.

25. Ans: C

Sol: Boiling point of EtOH more than EtCl , Me_2O , EtNH_2 & less than AcOH

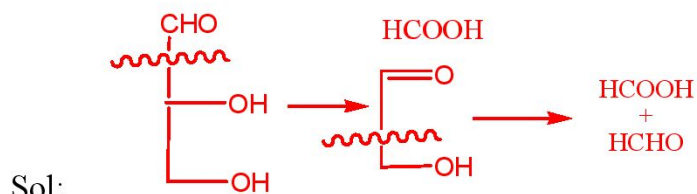
26. Ans: C

Sol: Semipinacol-pinacolone rearrangement

27. Ans: C

Sol: Divinyl ether is planar because all carbons are sp^2 hybridized

28. Ans: B



29. Ans: A

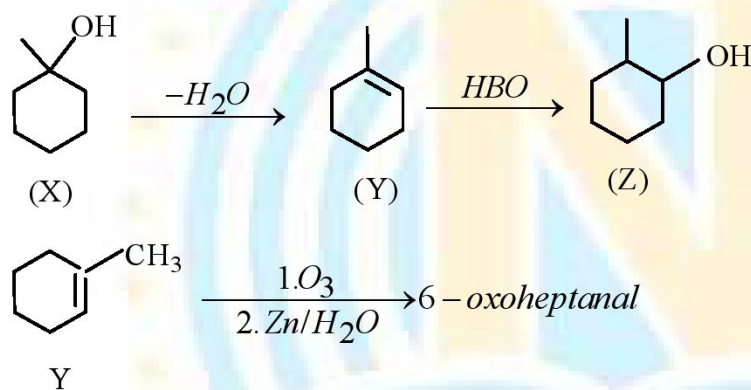
Sol: Sodium does not react with ethers

30. Ans: B

Sol: Phenol reacts with Phthalicanhdride in presence of H_2SO_4 to be phenolphthalein

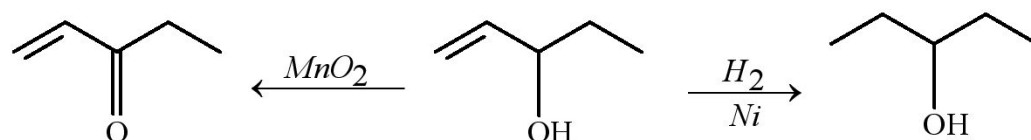
31. Ans: B

Sol:

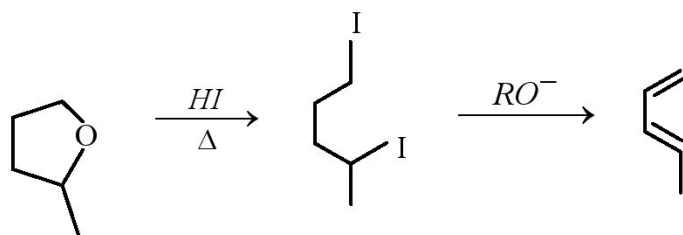


32. Ans: A

Sol:

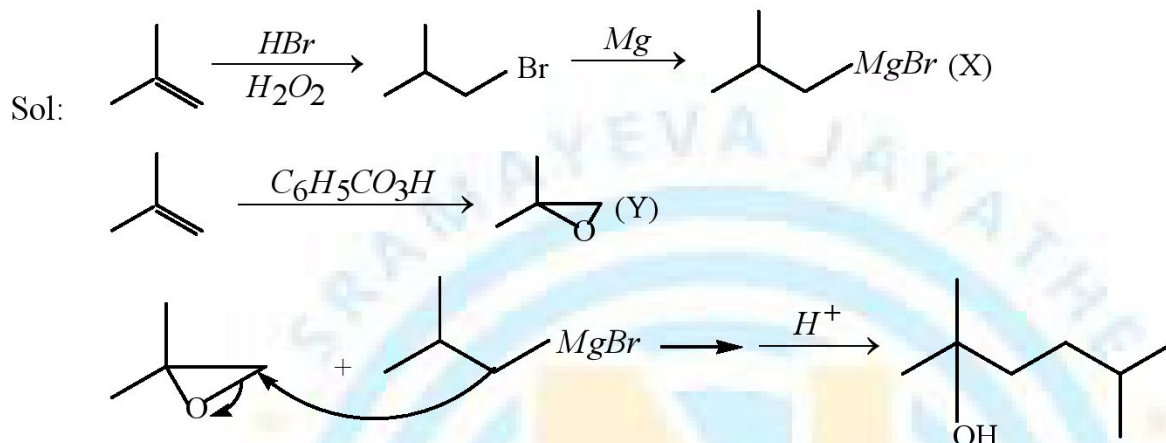


33. Ans: B

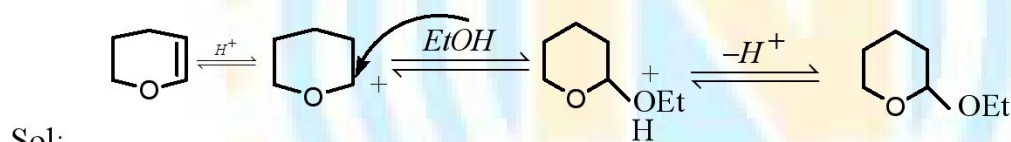


Sol:

34. Ans: B



35. Ans: C



Sol:

36. Ans: D

Sol: Only primary and secondary alcohols gives positive dichromate test

37. Ans: B

Sol: Order of basis strength is $-O^- < -C \equiv C^- < -CH = CH^- < -CH_2^-$

38. Ans: A

Sol: $C_4H_9OH \rightarrow C_4H_9Cl \rightarrow C_4H_9CN \rightarrow C_4H_9COOH$

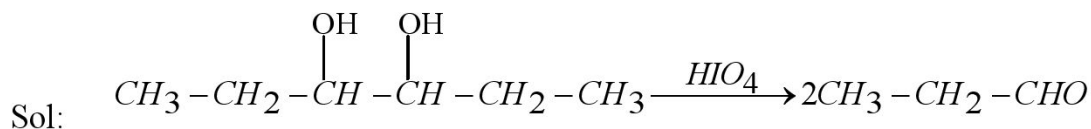
39. Ans: B

Sol: Aldehydes are oxidized to acids through formation of gem diols which occurs in aqueous medium

40. Ans: D

Sol: NaBH_4 reduce keto group but not ester, LiAlH_4 can reduce both

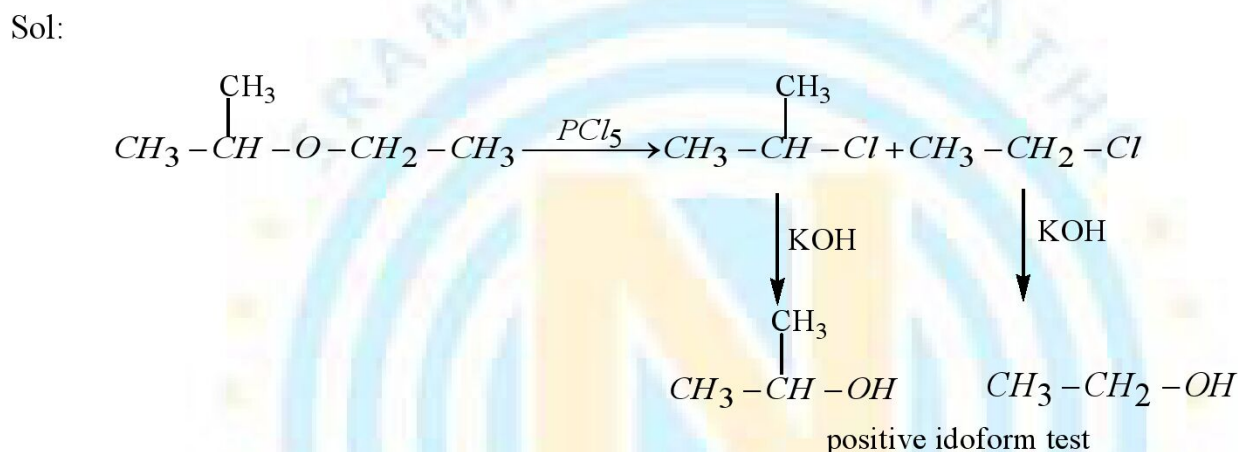
41. Ans: D



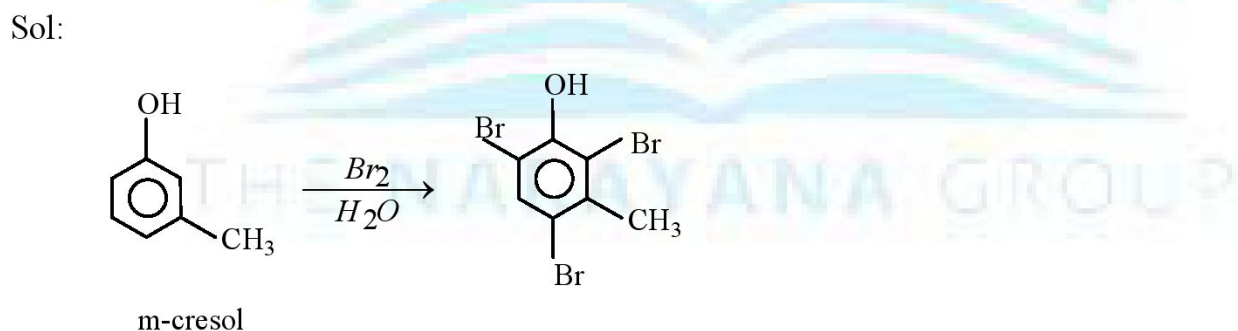
42. Ans: C

Sol: With hot Copper primary alcohol are oxidized to aldehydes and secondary alcohols are oxidized to ketones. Which can be distinguished by fehling solution

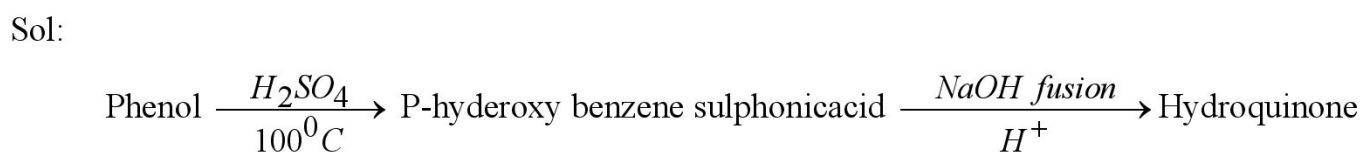
43. Ans: A



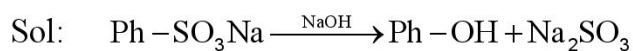
44. Ans: C



45. Ans: C



46. Ans: A

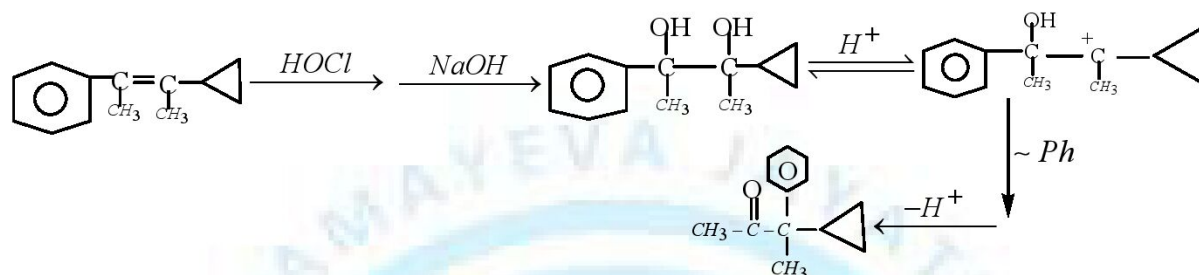


47. Ans: D

Sol: Claisen rearrangement

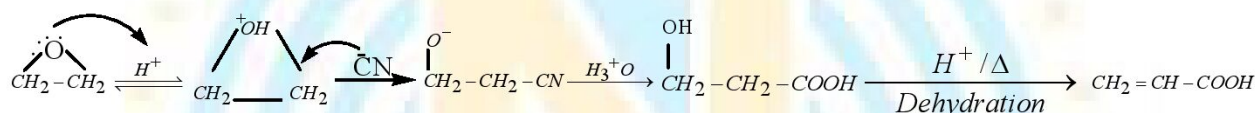
48. Ans: A

Sol:



49. Ans: A

Sol:



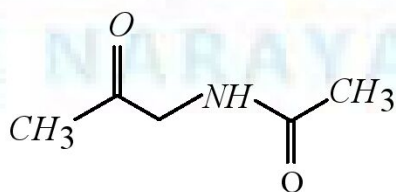
50. Ans: C

Sol: Reductive amination of aldehydes with primary amines gives secondary amines

51. Ans: A

Sol: 2-methyl propane2-ol undergo dehydration to give 2-methyl propane

52. Ans: B

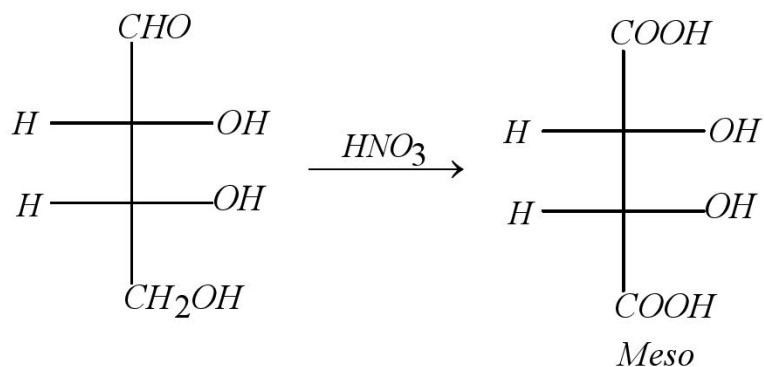


Sol: Product contain

keto group and amide group

53. Ans: D

Sol:



54. **Ans: B**

Sol: Absence of H-bonding in ethers

55. **Ans: C**

Sol: Second reaction is Hoffmann bromamide degradation

56. **Ans: C**

Sol: All are primary Amines

57. **Ans: A**

Sol: Absence of Hemiacetal group

58. **Ans: C**

Sol: Decarboxylation followed by Iodoform reaction

59. **Ans: D**

Sol: Aniline is soluble in dilute HCl

60. **Ans: C**

Sol: Vitamin B₁ is called Thiamine

61. **Ans: A**

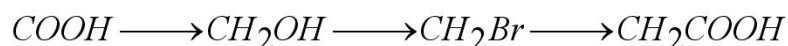
Sol: Amine undergo Acylation

62. Ans: D

Sol: Carboxylic acids liberate CO_2 with sodium bicarbonate

63. Ans: C

Sol: Following changes occurs in this transformation



64. Ans: A

Sol: Following changes occurs in this transformation



65. Ans: C

Sol: Only C_1, C_2 carbons are involved in the formation of Osazone

66. Ans: C

Sol: Vitamin D can be stored in our body

67. Ans: A

Sol: Secondary amines gives enamines with carbonyl compounds

68. Ans: B

Sol: This is Hoffman bromamide degradation which occurs with retention of configuration

69. Ans: C

Sol: Amides are less basic and sp^3 nitrogen is more basic

70. Ans: C

Sol: Intra molecular cannizaro reaction followed by esterification

71. Ans: C

Sol: Asprin is acetyl salicylic acid

72. Ans: B

Sol: Anomers differ at functional group carbon

73. Ans: B

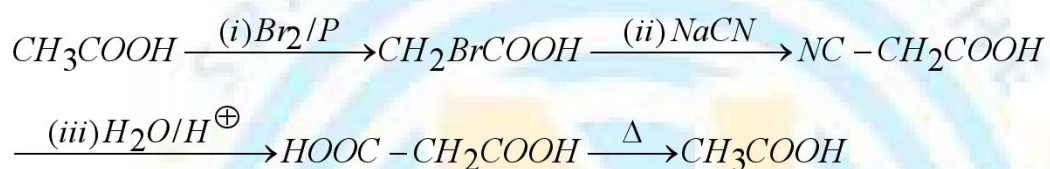
Sol: Absence of carbon in hydroxyl amine

74. Ans: A

Sol: Primary structure of protein is sequence of amino acids

75. Ans: A

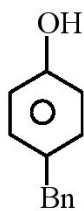
Sol:

NUMERICAL

76. Ans: 2

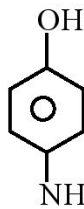
Sol: $\ddot{\text{N}} = \overset{+}{\text{N}} = \ddot{\text{N}}^-$

77. Ans: 1



Sol:

78. Ans: 8



Sol:

79. Ans: 5

Sol: Millimoles = $10 \times 10^{-3} = 10^{-2}$

$$\text{Moles} = 10^{-5}$$

$$\text{Number of molecules} = 6 \times 10^{23} \times 10^{-5} = 6 \times 10^{18}$$

$$\text{Surface area occupied by one molecule} = \frac{1.5}{6 \times 10^{18}} = 25 \times 10^{-20}$$

$$a^2 = 25 \times 10^{-20}$$

$$a = 5 \times 10^{-10} \text{ cm}$$

$$= 5 \text{ pm}$$

80. Ans: 3

Sol: $5.6'V'H_2O_2 = 1N$

$$10'V'H_2O_2 = \frac{10}{5.6} N$$

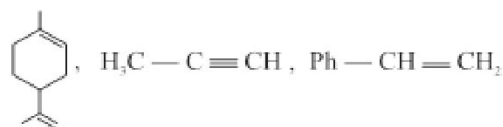
$$\text{mEq of } H_2O_2 = \text{mEq of } KMnO_4$$

$$\frac{10}{5.6} \times 336 = 0.04 \times 5 (n \text{ factor}) \times V$$

$$V = 3000 \text{ ml} = 3 \text{ litre}$$

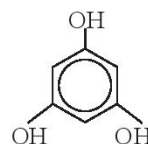
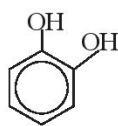
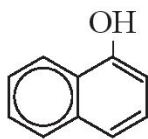
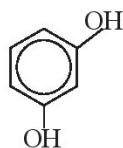
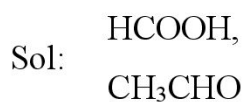
81. Ans: 5

Sol:



will produce CO_2 on oxidative ozonolysis.

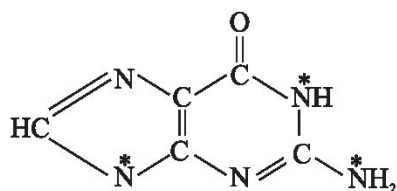
82. Ans: 6



Can reduce Tollen's reagent.

83. Ans: 3

Sol: The nitrogen's marked as * involves in resonance



84. Ans: 4

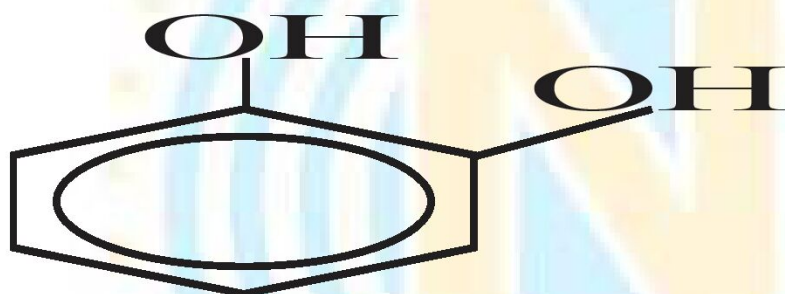
Sol: 1st, 2nd, 6th, 7th, will undergo aldol condensation

85. Ans: 4

Sol: Picric acid, sulfonic acid, formic and, nitric acid liberates CO₂ from NaHCO₃. Infact acids stronger then H₂CO₃ can liberate CO₂.

86. Ans: 2

Sol:



No. of Hyper conjugative structures=

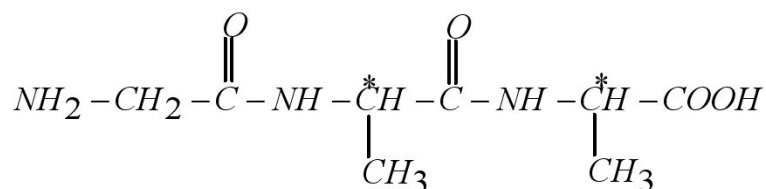


Refer Page No:-25.sec-1.9(A)

Book Name:-Solomon Fryhle

87. Ans: 3

Sol:



Glycyl alanyl alanine

88. Ans:5

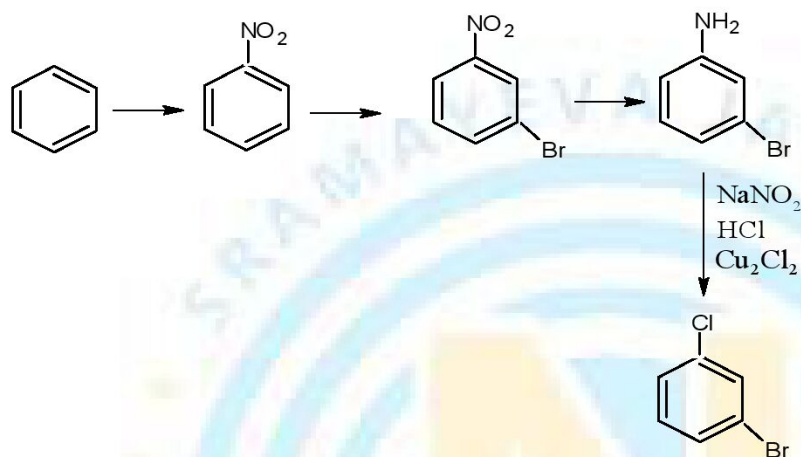
Sol. Only primary Amines can be prepared and R-X should give SN_2 so no vinyl and aryl halide can be used.

89. Ans : 1

Sol : only aromatic primary amines gives that test.

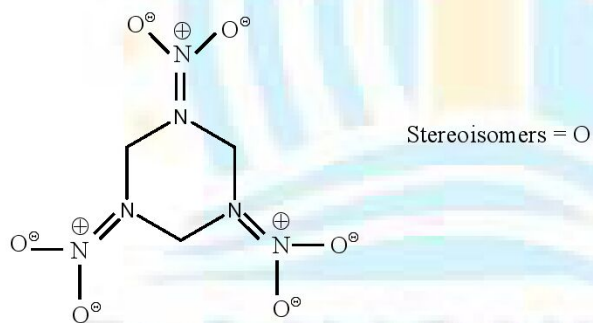
90. Ans: 3

Sol.



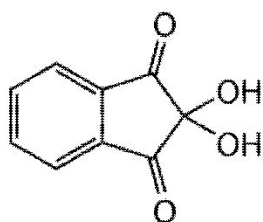
91. Ans: 0

Sol.



92. Ans: 8

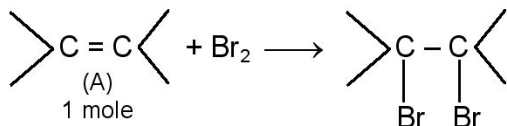
Sol: Conceptual, Modified question from NCERT.



93. Ans: 4

Sol: i) (A) gives monobromoalkane (B) $\rightarrow$ (A) is alkene

ii)



Since 2 gm Br_2 reacts completely with = 0.70 g of (A)

$$\therefore 160 \text{ gm } \text{Br}_2 \text{ reacts completely with} = \frac{0.70 \times 160}{2} = 56 \text{ gm of (A)}$$

$\therefore$ Molecular weight = 56

$$\text{C}_n\text{H}_{2n} = 56 \quad (\text{since compound is alkene})$$

$$12n + 2n = 56$$

$$n = 4$$

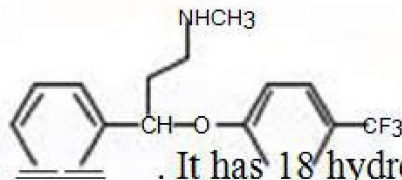
Hence (A) is C_4H_8

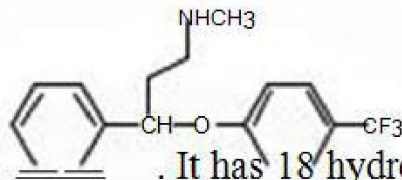
94. Ans: 4

Sol: In (ii), (iii), (iv) and (v) first compound is more reactive than the second compound.

95. Ans : 9

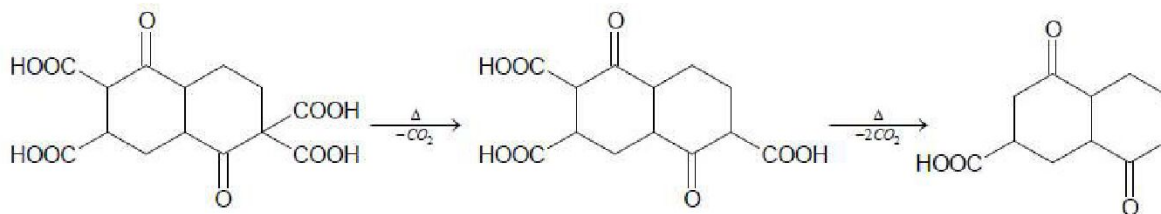
Sol:



The compound Prozac is . It has 18 hydrogens.

96. Ans : 3

Sol:



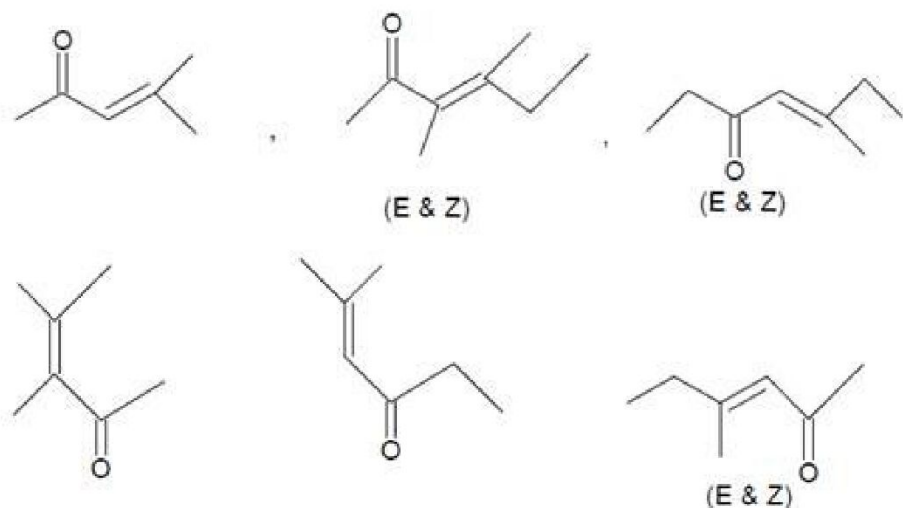
97. Ans : 2

Sol:

$\text{H}_3\text{C}-\overset{\text{O}}{\parallel}{\text{C}}-\text{H}$ and $\text{H}_3\text{C}-\text{CH}_2-\overset{\text{O}}{\parallel}{\text{C}}-\text{H}$ react with Fehling solution.

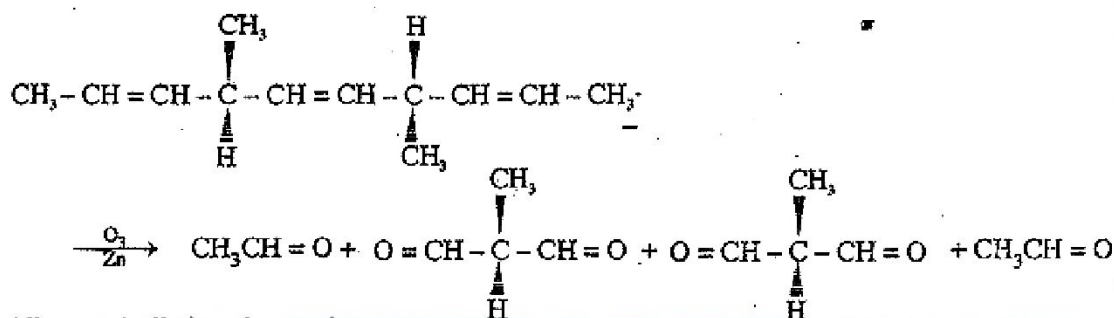
98. Ans : 9

Sol:



99. Ans: 0

Sol:



All are optically inactive products

No. Of optically active product = 0

100. Ans: 4

Sol:



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